

PROCESS EVALUATION FOR THE CLIMBER INITIATIVE ACTIVITIES IN KENYA



PROCESS EVALUATION REPORT

MAY 2024

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Acknowledgements

We are greatly indebted to the ClimBeR team who responded to our interviews and provided critical documents for the smooth flow of the process evaluation exercise. Special thanks go to Brian Mayanja for organising online engagements with the ClimBeR staff and other respondents beyond ClimBeR. We also appreciate the Key Informants from the local implementing partners and/or collaborating institutions such as the Equity Bank, Family Bank, APA Insurance, CEMIRIDE, MEDIAE, KIPPRA, AUC, IGAD, JKUAT, adelphi, NDMA, RECONCILE and the Ministry of Environment, Climate Change and Forestry. We cannot forget the farmers who generously gave their time to respond to the evaluation questions, either as individuals or in groups.

Abbreviations and Acronyms

AWARE	Early Warning, Early Action, Early financing
FGD	Focus Group Discussion
GSE	Gender and Social equity
GCF	Green Climate Fund
IGAD	Inter-Governmental Authority on Development
KII	Key Informant interview
LNOB	Leave No One Behind
TWG	Technical Working Group

Executive Summary

In response to the threats of climate change, Alliance of Bioversity International and CIAT and its partners launched the CGIAR Research Initiative on Climate Resilience (ClimBeR) in 2022 in Nairobi, Kenya, covering 6 countries (Guatemala, **Kenya**, Morocco, the Philippines, Senegal, and Zambia). The aim of ClimBeR is to simultaneously transform food, land, and water systems in a climate crisis so as to enhance the resilience of the smallholder production systems. In 2023, ClimBeR initiative commissioned this process evaluation exercise with two main objectives: (1) To assess the adherence to implementation plan, noting any deviations & corrective measures, and (2) To demonstrate how ClimBeR **impacts are emerging**, highlighting the **drivers of and barriers to uptake**.

The process evaluation used a blend of qualitative and quantitative approaches. The qualitative module involved Key Informant Interviews (KII) with ClimBeR staff and country-level implementing partners and focus group discussions (FGDs) with implementing farming communities. Quantitative module involved household interviews in the communities where the Risk-Contingent Credit (RCC) innovation is being implemented. The two modules were supplemented with literature review.

The evaluation found that ClimBeR had made tremendous progress, sticking to its theory of change (ToC) and the activities programmes to a large extent. While there were slight delays in implementation of some of the planned activities, the ToC remained unchanged. More important, even for the activities that were behind schedule, the responsible staff confirmed that they would still accomplish them within the project life. The main milestones by Work Packages include the following:

Work Package 1

1. Bundled financial products are already developed and socialised among the smallholder farmers, taking into account gender and social equity. The products have been rolled out and taken up by some farmers. It is notable that male farmers remain more likely to take up the bundled services than their female counterparts. Similarly, households with better access to farm labour are more likely to take up the products.
2. Farmers are receiving more useful agro-climatic information services, courtesy of partnership between ClimBeR and the private company disseminating the information. More important, the farmers with access to agro-climatic information services are more likely to take up the bundled products.
3. Derisking of the smallholder production systems enhances access to agricultural finance, thus justifying the bundling of the two products.

Work Package 2

1. Climate Security Sensitivity Tool (CSST) developed and used to assess the peace responsiveness of projects and programmes.
2. Climate Security Index (CSI) methodology already developed.
3. Climate Security Observatory (CSO) already developed and functional.
4. Training course on climate, peace and security in agriculture developed, validated, and piloted in one county.
5. Contributions to mainstreaming Climate, Peace, and Security into Kenya's National Climate Change Action Plan III (Climate Security Actions), IGAD Climate Change Adaptation Strategy, and Africa Climate Security Assessment lead by adelphi.

Work Package 3

1. Policy landscape assessment has been done and simulation scenarios are being discussed.
2. Technical Working Group (TWG) has been formed and multiple meetings have led to progress.
3. Busia County engagement (see pg 28).
4. Activities are a little behind schedule, but now that most of our policy engagement work is focussing on Kenya, we are effectively on track.

Work Package 4

1. Leave No One Behind (LNOB) indicators have been developed but are yet to be socialised with the Government of Kenya.
2. Polycentric governance model has been tested in Kajiado to assess how local communities can take ownership of adaptation strategies.
3. Work on *Climate Smart Governance dashboard* and Early Warning, Early Action, Early financing (AWARE) platform are yet to begin in Kenya. These will need to be fast-tracked to be completed in time.

Gender and Social Equity (GSE)

1. Voices of change trained and given appropriate platform.
2. Gender and social equity indicators and tools developed and used.
3. Gender-sensitive training materials developed and used under RCC.
4. Gender and social equity markers and narratives developed.

Climate Finance

1. Climate finance credit scoring tool developed and adopted.
2. Finance and climate literacy training information provided.
3. Climate security investment plan methodology developed.
4. GCF concept developed for Turkana, Baringo, Kilifi, Narok and West Pokot.
5. CSAIP developed through AICCRA, Integrated into GCF proposal, awaiting signature of the Principal Secretary (PS) of the Ministry of Agriculture and Livestock Development.

Early indicators show that ClimBeR is on track to achieving the intended impacts. A number of work packages have moved from activities to outcome level while some have reached the output level. However, a few interventions are still at the activity level. The success is attributed to a participatory approach and focus on the ToC which have led to systems thinking, collaboration with partners and recognition of diversity of livelihood pathways. It is also attributed to keen monitoring and supervision of activities through output-based contracts and sub-contracts, periodic reporting, working on shared platforms, regular meetings within and across teams and review meetings. Despite the milestones realised, special attention on the activities that are behind schedule will be required in the final year of the Initiative. The activities will need to be fast-tracked (where possible) and be toned down to light touch as increased attention is given to high impact activities.

1. Introduction

1.1 Background

Climate change and variability, often manifesting in increased frequency and intensity of extreme weather events and incidence of pests and diseases, leads to large financial losses (Shakhovskoy & Mehta, 2018). For example, the droughts between 1980 and 2013 in sub-Saharan Africa (SSA), affected about 360 million people and occasioned losses of about US\$ 31 billion (FAO, 2015). In Kenya, for example, between 2008 and 2011, droughts resulted in losses of about US\$ 11 billion (FAO, 2015). In Malawi, on the other hand, the drought of 2016 led to losses estimated at US\$ 400 million (Reeves, 2017). In Mozambique and Tanzania, every five years, extreme weather leads to 61% and 72% of crops losses while pests lead to 32% and 38% of crop losses, respectively (Anderson & Ahmed, 2016). Drier areas of East Africa, particularly, have suffered from the cumulative effects of failed rainy seasons, and increased frequency and intensity of droughts, which have lowered animal growth rates and productivity in pastoral systems, adversely affecting food security (WMO, 2020, 2021). In other areas, high precipitation has led to flooding. In 2019 and 2020, for example, high precipitation and above normal vegetation growth favoured breeding of desert locusts, with swarms migrating from one country of East Africa to another, destroying crops and pasture. Largely, it is anticipated that climate change will lead to burgeoning poverty and food insecurity through disruption the growing seasons, reduction of areas suitable for agriculture and deterioration of agricultural yields (Connolly-Boutin & Smit, 2016; Spinoni *et al.*, 2020).

For a region like East Africa where farmers are largely subsistent and heavily dependent on rain-fed agriculture, adverse effects of climate variability and change are likely to result in increased food insecurity, reduced and erratic agricultural incomes, and increased poverty (AGRA, 2019; FAO *et al.*, 2018). To cushion themselves, households, over the years have resorted to a myriad of strategies such as migration, income diversification and adoption of improved and more appropriate technologies (Babatunde & Qaim, 2010; Burney & Naylor, 2012; Karamba *et al.*, 2011), and use of climate information for farm decisions (Birachi *et al.*, 2020; Radeny *et al.*, 2019). While these strategies are associated with some positive outcomes, they are largely sub-optimal, being marginal rather than transformative responses to climate change (Gbegbelegbe *et al.*, 2018; Kristjanson *et al.*, 2012; Nyasimi *et al.*, 2017). This is partly because of the increasing incidence and severity of climate-related risks and extremes, and partly due to the low adaptive capacity of the smallholder farmers. It is against this backdrop that **CGIAR Research Initiative on Climate Resilience** (ClimBeR) is being implemented— to deliver science and innovation to simultaneously transform food, land, and water systems in a climate crisis. ClimBeR is being implemented in six countries: Guatemala, **Kenya**, Morocco, the Philippines, Senegal, and Zambia. The Initiative aims at enhancing the resilience of smallholder production systems to episodes of drought, flooding, and high temperatures associated with climate change.

The anticipated outputs/outcomes by the end of 2024 include the following:

1. Bundled ClimBeR climate services reaching at least 300,000 vulnerable farmers in the countries of implementation, and at least 30% of the beneficiaries being women.

2. Policy makers and international agencies having adopted ClimBeR products to influence at least nine policies or to strengthen agricultural resilience, at least two of which should reduce agriculture-related climate security risks.
3. Initiative partners having been enabled or supported to invest US\$ 30 million targeting women, youth and other vulnerable groups, leading to building of the resilience of these groups.

1.2 About the Process Evaluation

The CGIAR Initiative on Climate Resilience (ClimBeR) is implemented in six countries which include Guatemala, Kenya, Morocco, the Philippines, Senegal, and Zambia. In Kenya, ClimBeR is implemented with local partners including the public sectors, civil societies, and the private sector. The aim is to ensure systemic transformation for enhanced resilience, productivity, and equity. It builds on CGIAR's previous work on agricultural research for development, and more particularly on participatory action research by CGIAR Research Program on Climate Change, Agriculture and Food Security (CCAFS). ClimBeR activities in Kenya are spread across six broad areas:

1. De-risking (Work Package 1): That is, reducing risks for producer livelihoods and in value chains. This is because smallholder agriculture in Kenya is largely rain-fed and highly susceptible to climate-induced risks, both in production and post-harvest management. Because these risks are not clearly understood, ClimBeR researchers use *agricultural risk management, climate-smart innovations, and digital services* to reduce them. Among the planned key innovations are: (i) Agricultural risk management strategy through Risk-contingent credit (RCC)/bundled products; (ii) Agro-climate information services through digital tools to enable climate-informed investments and promote uptake of climate smart agriculture (CSA); and (iii) Production of farming system typology with nutrition lens for better targeting of financial investment.
2. Climate security (Work Package 2): Building production system resilience while recognising the links among climate, agriculture, security, and peace. This component, therefore, involves understanding *climate-related security risks and identifying pathways to peace*. This is because climate change may escalate conflicts over resources among communities which are already vulnerable. To mitigate this, it is important to generate *robust, local-level policy relevant evidence* about the possible risks. Key innovations include: (i) Climate Security Observatory (CSO) to qualify and quantify climate security at regional, national and sub-national levels; (ii) Climate Security Index (CSI) to monitor climate security risks; (iii) Climate security proofing guidelines for localised climate security assessments to inform planning and programming, and for conflict-sensitive climate adaptation and agricultural policies; and (iv) Governance and policy for climate security tool to guide national and sub-national policy-making for synchronised climate, agriculture, and peace programming to enhance effectiveness of climate adaptation.
3. Policy pathways (Work Package 3): Developing adaptation instruments to shape policy and investments. In Kenya, ClimBeR works with partners, policy makers and stakeholders to develop the integrated Future Estimator for Emissions and Diets (iFEED), an evidence-based framework to identify *policy pathways to climate-resilient, food- and nutrition-secure future* for Kenya. The iFEED is expected to enable policymakers to make decisions that deliver nutrition security while enhancing farmer climate resilience.

4. Governance for resilience (Work Package 4): ClimBeR researchers and partners are *building and strengthening networks for co-designing, co-developing, and co-implementing adaptation interventions and coherent policies* (Multiscale governance for transformative adaptation). This will ensure that the needs and priorities of local communities and best practices are embedded in the interventions and policies. To achieve this, ClimBeR and partners are developing: (i) Bottom-up polycentric multi-scale governance model for climate adaptation, linking local-level adaptation with national level policy processes; (ii) Integrated framework with ‘Leave No One Behind’ indicators; (iii) Climate Smart Governance Dashboard that captures global level data and community needs for evidence-based policy/decision-making; (iv) AWARE Platform for coordination of early warning, early action and early finance in response to extreme climate events; and (v) Capacity building/technology transfer for multiscale adaptive management (champions of change).
5. Climate finance: This is a cross-cutting theme, contributing to Work Packages 1, 2, 3 and 4. ClimBeR researchers and partners are developing mechanisms to *increase access to finance by farmers at local level, and to enable policy makers to identify new funding opportunities and mobilise additional resources*. The target innovations are: (i) co-designing, with financial institutions, *commercially viable risk-informed financial service products* (microcredit/microfinance business models) for farmers; and (ii) Finance for peace—climate security investment plans. Mainstreaming GSE in its work streams, Climate Finance contributes to the other work packages as summarised in Table 1.

Table 1. Contribution of Climate finance Work package to other Work Packages

<i>Work Package</i>	<i>Elements that CF contributes to</i>
WP 1-De-risk	1. Agro-climatic informed Innovative financial mechanisms combining credit, savings, and insurance to build resilience (RCC)
WP 2-Climate Security	2. Climate risk profiling system to de-risk finance and investments
	1. Humanitarian and resilience building investments.
	2. Innovative finance mechanisms for peace benefits
WP 3-Policy Pathways	1. CSA Investment Planning
	2. Public-private blended finance mechanisms for scaling resilience building interventions
WP 4-Governance 4	1. Financial and investment indicators
Resilience	2. Institutional structures (e.g. cooperatives, savings groups) for financial literacy and access to finance

6. Gender and social equity: Climate risks are not likely to impact people equally. Women, youth, and other disadvantaged groups are more likely to be impacted more due to their low adaptive capacities. Thus, ClimBeR is intentionally integrating gender and social equity in pursuing a climate-resilient and nutrition-secure future. The approach is to design interventions that cater for the diverse needs of farmers and devoid of maladaptation. To achieve this, this theme seeks to: (i) Develop gender and social equity indicators; (ii) undertake capacity building of the youth leadership on climate change and negotiations and document the voices of change; (iii) Developing and testing gender and social equity tools; (iv) Designing gender-sensitive training materials; (v) Documenting social equity narratives; and (vi) Designing social equity markers. Although the work stream has

specific activities and deliverables that it accounts for directly, it is important to note that GSE is cross-cutting, its deliverables feeding into other work streams as summarised in Table 2.

Table 2. Contribution of GSE to other Work Packages

<i>Work Package</i>	<i>Elements that GSE contributes to</i>
WP 1-De-risk	1. Incorporating gender and social equity in assumptions of climate risk profiling system
	2. Understanding barriers to uptake of climate information services
WP 2-Climate Security	1. Identifying & prioritizing the most vulnerable to ensure peace and security in a climate crisis.
	2. Attention to imbalances in access and distribution of resources, processes, & representation
WP 3-Policy Pathways	1. Incorporating gender and social equity in assumptions for policy pathways
	2. Embedding gender and social equity in participatory workshops, future visioning, vulnerability mapping
WP 4-Governance 4 Resilience	1. Multi-stakeholder approach to farmer/community empowerment and institutional responses to climate shocks
	2. Indicator framework to design local investments with a focus on disadvantaged groups

Implementation of the ClimBeR initiative in Kenya started in 2022, with an inception meeting being held on the 21st June, 2022, bringing together 41 regional and national organizations. By the time the process evaluation was commissioned, ClimBeR activities had been implemented for over one and a half years and the program was moving into its second phase of implementation. This made it important to reflect on what had been implemented to draw lessons on what had worked, what had not worked, what needed to be modified, what needed to be maintained and what may have needed to be dropped, to optimize the program outcomes. This reflection was also important in taking stock on the overall progress towards the intended outputs, outcomes, and impacts. The evaluation was based on ClimBeR activities in Kenya.

Process evaluation relies on impact pathways that trace how project *inputs/resources* are used in various *activities* across different intervention areas to deliver *outputs* that could contribute to *outcomes* and eventually to long-term *changes or impacts*. See Figure 1 for a generic illustration of impact pathway.

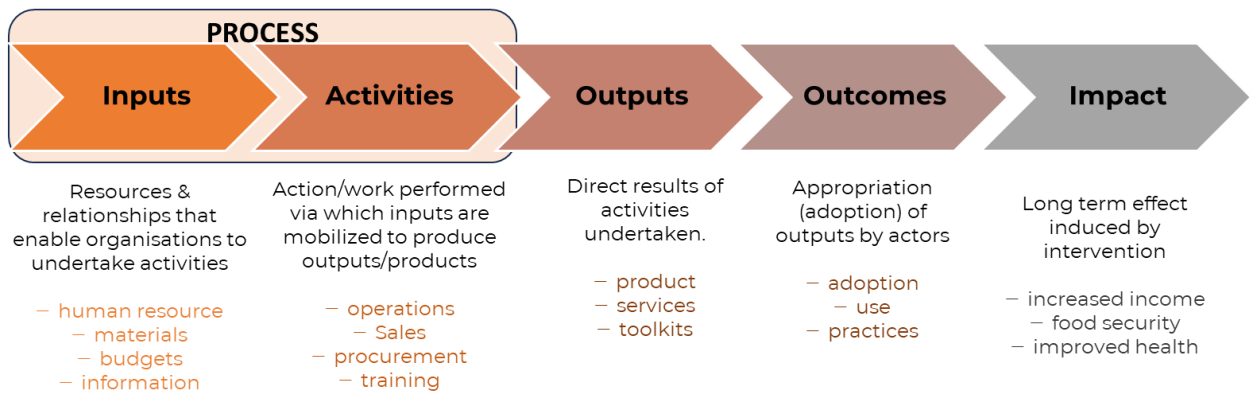


Figure 1: Generic illustration of impact pathway

Although some interventions had gone beyond the output stage, the process evaluation covered mainly the first 3 stages of impact pathways as described below:

1. Inputs which may include intervention design, financial, technical, and managerial investments, human resources, etc.
2. Processes which include the system in which things are done such as quality management, mobilization of stakeholders, training preparation, sequencing of activities, participatory action research, etc.
3. Activities such as the individual actions (tasks and sub-tasks) that must be undertaken for the success of the initiative (e.g. training, collecting data, convening meetings).
4. Outputs which are the materials or events or products of the intervention (e.g. seed, policy, regulation, plan, toolkit, digital app, insurance product, etc.).

The evaluation paid attention to understanding the processes or systems within which activities were undertaken. Additionally, the evaluation sought to understand barriers to delivery of outputs that should be addressed and enablers that should be emphasized going forward.

The overarching objective of the evaluation was, therefore, to examine the extent to which the project had adhered to the strategy as laid down in the logic model and if the programme was delivering intended outputs. The specific objectives were twofold:

1. To assess the extent to which implementation of the ClimBeR Initiative in Kenya adhered to the plan and identify any deviations and the corrective measures taken.
2. To demonstrate how ClimBeR impacts were emerging from the inputs and the activities, highlighting the drivers of and barriers to participation and uptake.

The first objective focuses on the *implementation process* and its *effectiveness*, including the *relevance* of implementation approaches to achievement of intended impacts. Among other things, this objective sought to understand if planned activities and approaches had been applied or if there were challenges and contexts that necessitated changes in plans. If there had been changes, this objective sought to provide an understanding of how such changes could have affected the achievement of intended goals and if changed plans and activities would still address the needs of target groups and partners. This objective was addressed by answering the following questions:

1. What approaches were planned and/or used in the implementation of activities? How effective were these approaches and how were they contributing to the progress towards the intended impact?
2. What mechanisms were planned and/or used in the monitoring and supervision of ClimBeR activities and how effective were they?
3. Was ClimBeR appropriately responding to the needs of the partners?
4. What changes had occurred in the operational landscape in Kenya, and what adjustments were required in the ToC in response to the changed context?

The second objective sought to demonstrate if and how ClimBeR activities were on course to delivering impacts. This objective called for a review of inputs and activities to understand if they were yielding outputs that were indicative of a journey to success, including identification of enablers and barriers to wider uptake of ClimBeR's innovations. The evaluation exercise for this objective was guided by the following questions as provided in the terms of reference:

1. How was ClimBeR initiative being managed and how had this style supported researchers, policymakers, and development practitioners to work *collaboratively*, embrace *systems-thinking*, and recognize the *diversity of livelihood pathways* leading to climate resilience?
2. Were ClimBeR products attracting private sector investment and how far had they attracted investments in climate finance? By whom and how well were the associated activities implemented?
3. How was ClimBeR helping stakeholders to embrace the concept of gender and social equity?
4. How were the WPs and crosscutting themes linked?
5. How was ClimBeR working together with other relevant initiatives?

Finally, the evaluation exercise drew lessons from the ClimBeR management style, implementation process, and associated approaches to provide appropriate recommendations for deepening ClimBeR's intended impacts.

2. Evaluation approach

2.1 The overall design

As explained in section 1, the process evaluation leverages the impact pathways approach and reconstructs chain of events from one probable cause to another in order to assess how particular events or activities could have influenced realized outputs and observed outcomes. While quantitative analysis based on household surveys can reveal if changes have occurred and the extent of those changes, process evaluation provides insights into how and why the identified changes occurred. Among other things, process evaluation establishes evidence of activities undertaken and if respective activities resulted in relevant outputs. Moreover, process evaluation seeks evidence of achievements of outcomes and substantiates the significance of hypothesized contributing factors relative to other possible contributors to outcomes. Process evaluation can, therefore, show whether a change occurred in the

process of project or program implementation, while also identifying why and how the change occurred (Punton & Welle, 2015). This enables organizations to identify their contributions to change and thus demonstrate accountability for results. Additionally, process evaluation can enhance performance by providing lessons on what works and why. The method is suitable for analyzing policy influence and capacity development where change may be due to many different influences and therefore attribution is contestable.

In examining the influence of project activities on supposed changes, process evaluation relies on a set of formal tests to assess causation. The tests focus on evidence with different kinds of probability, thus either confirming or eliminating potential explanations of occurrence. Evidence derived from empirical observations, in combination with knowledge of the context, timeline of occurrence and the manner in which facts emerge help to unpack the causal mechanism that explains what and how a cause leads to an outcome (Befani & Mayne, 2014).

The causal mechanism is composed of “parts of entities” (*people, organizations, systems, etc.*) that engage in activities (*pre-testing, researching, campaigning, etc.*), with each part of the mechanism giving rise to a subsequent part (Beach & Pedersen, 2019). This causal chain that links event A with outcome B forms an explanation or a hypothesis to be tested. The tests are applied to all possible explanations (hypotheses) on how a particular change might have occurred, thus confirming, or eliminating alternative explanations accordingly.

Since more than one causal chain may contribute to an effect, it is important that one identifies all possible explanations of observed or potential outcomes before subjecting each to the set of tests outlined below. In Table 3, we explain the set of tests that hypotheses are subjected to.

Table 3. Hypotheses Tests

		Sufficient to affirming causation	
		No	Yes
Necessary to affirming causation	No	1. Straw in the wind test	3. Smoking gun test
		Passing: affirms relevance of hypothesis but does not confirm it Failing: suggests hypothesis not relevant but does not eliminate it. Implication for rival hypotheses: None	Passing: confirms hypothesis Failing: does not eliminate hypothesis. Implication for rival hypotheses: None
	Yes	2. Hoop test	4. Double decisive test
		Passing: affirms relevance of hypothesis but does not confirm it Failing: eliminates it. Implication for rival hypotheses: None	Passing: confirms hypothesis Failing: falls short in establishing necessity and/or sufficiency. Implication for rival hypotheses: eliminates all other hypothesis

Further illustration of the 4 tests with an example

The *straw in the wind* test provides information that may favor or call into question a given hypothesis. Such information on its own is, however, indecisive. The information is **neither necessary nor sufficient** to affirm or reject a hypothesis. Passing a straw in the wind test therefore supports a particular hypothesis but does not rule it in or out. Neither does the test have implications for competing hypotheses. Take the example of the risk-contingent credit,

If the Equity Bank offers risk-contingent credit (RCC) and the ClimBeR initiative is listed as working on RCC innovation in Kenya, the mere operational presence of ClimBeR in Kenya is not decisive piece of evidence. However, work of ClimBeR on RCC issues in Kenya might increase plausibility that ClimBeR has influenced the development of RCC that is currently offered by Equity Bank.

The *hoop* test can eliminate alternative hypotheses but does not provide direct supportive evidence for a hypothesis that has not been eliminated (Bennett, 2010). The test provides **necessary but not sufficient** criterion for accepting an explanation. Therefore, explanation/hypothesis must “jump through the hoop” to remain under consideration. However, passing the hoop test does not affirm a hypothesis. For instance:

If ClimBeR is listed by Equity Bank (or one of the affiliate CG centers, e.g., ABC) as one of their partners and we know that ClimBeR works in developing appropriate RCCs, then the identification of ClimBeR as a partner of Equity Bank passes the hoop test. Otherwise, operational presence of ClimBeR in Kenya alone does not necessarily mean that they have influenced the development of the RCC product offered by Equity Bank.

The *smoking gun* test on the other hand offers **strong support** for a given hypothesis. Nevertheless, failure to pass the smoking gun test does not eliminate the hypothesis. Thus, the test provides **sufficient but not necessary** conditions for confirmation of a hypothesis. Continuing with our example from the hoop test:

If Equity Bank provides testimonies that they have been collaborating with ClimBeR and other partners in developing their RCC product currently on offer, then the testimonies are sufficient evidence of causality. However, such testimonies from Equity Bank do not invalidate other hypotheses. Neither does the absence of such testimonies invalidate the hypothesis that ClimBeR may have influenced the development of Equity Bank’s RCC currently on offer.

Finally, a *double decisive* test provides **necessary and sufficient** evidence for accepting a hypothesis and eliminates other competing hypotheses. Moreover, any hypothesis that fails the double decisive test is eliminated. For example,

If Equity Bank gives testimonies that their collaboration with ClimBeR in developing the RCC product is “solely” responsible for RCC product that they have on offer, then the evidence is doubly decisive.

It is noteworthy that in development work where there are many factors at play, *double-decisive* evidence is rare (Van Evera, 1997). However, the hoop test applied in combination with the smoking gun test accomplishes the same analytical goal.

Since the application of these tests imply comparing alternative explanations, it is important that the analysis begins with several possible alternatives. Such explanations need to be drawn from the wider body of evidence. Thus, besides reviewing the project ToC to derive impact pathways for respective outcomes, we also considered several theoretical bases that may not be apparent in the project ToC.

2.2 Execution of the study

Following Bennett (2010), we implemented the process evaluation in 5 steps:

Step 1. Identify changes to be explained. The changes were extracted from project documents including the proposal, ToC/result framework/log-frame and project progress reports. The focus was on the intended outputs and/or outcomes which are displayed by the Work Package in Table 4. Further engagement with ClimBeR project team (Work Package leads) confirmed the extracted information.

Table 4: Outputs/Outcomes of ClimBeR work streams

Work Packages	Outputs
Work Package 1	1. Agricultural risk strategies 2. Climate information services 3. Climate risk profiling 4. Diverse farming system typologies
Work Package 2	5. Climate Security Observatory (CSO) 6. Climate Security Index (CSI) 7. Climate Security Proofing Guidelines 8. Climate Security Policy Tool 9. Climate Security Sensitivity Tool (CSST) 10. Climate Security Investment Plan (CSIP)
Work Package 3	11. Policy pathways for climate-resilient, nutrition-secure & socially equitable futures
Work Package 4	12. Bottom-up polycentric governance model for climate adaptation. 13. Integrated framework with “Leave No One Behind Indicators”.
Climate financing	14. Climate smart governance dashboard 15. AWARE platform 16. Capacity building/technology transfer for multi-scale adaptive management (champions of change). 17. Microfinance business models 18. Finance for peace: climate security.
Gender & Social Equity	19. Support to governments in accessing climate funds 20. Gender & social equity indicators

21. Voices of change
 22. Gender & social equity tools
 23. Gender-sensitive training materials
 24. Social equity narratives
 25. Social equity markers
-

Step 2. Establish evidence for the change. This involved a review of project M&E documents, household survey, and key informant interviews (KIIs) with project staff to reveal:

1. What was considered the most significant changes.
2. The evidence and indicators of such changes and how measurable the changes were.

Step 3. Document the process leading to the change. From the identified changes and their associated indicators, we reconstructed the process to show how and explain why certain things happened in the process of realizing the respective changes. This process involved developing **impact pathways** to show how different interventions and activities could have led to identified changes/outcomes and impacts. This process helped in building coherent hypotheses on how project activities may have contributed to the identified changes. A review of the ToC with the project team helped in:

1. Developing narrative of activities that had been carried out.
2. Identifying tangible outputs that had been delivered.
3. Determining the resulting intermediate and eventual outcomes.
4. Identifying the external events that may have affected changes.

Further KIIs and/or FGDs with project partners and participants were used to systematically assess and document activities derived from workplans, identifying the critical ones that may have tipped the scale towards achieving the realized outputs. At the core of this step was the identification of causal stories to explain how some changes or outputs may have come about.

Step 4. Establish alternative causal explanations. Alternative explanations provide a set of different hypotheses of how and why changes came about. This is also called “*process induction*” and involves exploratory, inductive research to identify alternative explanations for the identified changes. A careful review of the ToC and the associated impact pathways, evidence from statistical analysis of household survey, testimonies from qualitative surveys (FGDs, KIIs, etc.) and literature review provided a comprehensive documentation of diverse explanation for the identified changes. For each alternative hypothesis, the documentation process outlined in step 3 was applied.

Step 5. Assess the evidence for each causal explanation. This involved applying the gathered data/information to verify the process and the extent to which the established hypotheses were supported by available evidence. Each developed hypothesis was subjected to the evidence needed for its confirmation or invalidation. The review process considered sequence and structure of events to establish if a given event triggered certain responses. During this process, each of the developed hypotheses was subjected to the 4 tests outlined earlier. We summarize the five steps in Figure 2 which also outlines the sources of information, and the data collection tasks (methods) used.

Steps	Information source	Tasks
 Step 1: Identify changes to be explained	- Project documents – proposal, ToC, result framework, progress report - Project staff	- Literature review - KIIs - FGDs
 Step 2: Establish evidence for change	- Project documents - Interaction with project staff, partners and beneficiaries	- Desk review of project M&E documents - Qualitative surveys – KIIs, FGDs, etc. - Quantitative household surveys
 Step 3: Document change pathways	- Project ToC or result framework - Testimonials - Workplans	- KIIs with project implementers - KIIs with project partners - FGDs with beneficiaries - Desk reviews
 Step 4: Establish alternative causal explanation	- Project ToC/result framework - Literature review - Testimonials - Surveys	- Desk review - Inductive analysis - KIIs with project staff - FGDs with project beneficiaries - Analysis of quantitative survey data
 Step 5: Assess evidence for each causal explanation	- Surveys - Reviews	Comparative hypothesis analysis

Figure 2. The 5-steps in process evaluation

2.3 Data collection

As outlined in section 1.2, the first 3 stages of the impact pathways – *inputs*, *activities* and *outputs* are the *drivers of impact*. Outputs provide good measures of success since they provide evidence that inputs and activities have yielded intended results, which when adopted by actors can lead to desired outcomes and impacts. Across the Work Packages, each output has indicators of success or milestones that show progress as captured in the ToC. These indicators include:

1. Number and diversity of partnership agreements signed.
2. Number of innovations developed such as RCC products, governance models, etc.
3. Number of men and women farmers reached by innovations.
4. Number of user/partner training workshops organized.
5. Knowledge products developed such as policy/technical briefs, blogs, webinars, peer reviewed articles, country factsheets, country framework documents, etc.
6. Consultation workshops organized.
7. Field research activities (fieldwork, surveys, etc.) undertaken.
8. Databases

These indicators informed the type and source of data collected.

In Table 5, we present some of the indicators of success and the relevant data sources or collection methods that were applied to assess progress towards achievement of objectives, organized by objectives and the associated evaluation questions.

Table 5: Indicators and data sources by evaluation questions

Evaluation questions	Indicators of success	Data sources/collection methods
Objective 1: To assess the extent to which implementation of the ClimBeR Initiative adhered to the plan and identify any deviations and the corrective measures taken.		
What approaches were planned and/or used in the implementation of activities? How effective are these approaches and how are they contributing to the progress towards the intended impact?	1. Adherence to/changes in work plan – processes, activities, etc. (<i>relevance of planned activities</i>) 2. Extent of achievement of planned results/outputs - indicators (<i>workshops, trainings, knowledge products - policy briefs, peer reviewed articles, etc.</i>)	1. Desk review of project documents – proposal, workplan, technical reports (2022 & 2023) 2. KIs with project implementation team
What mechanisms were planned and/or used in the monitoring and supervision of ClimBeR activities and how effective have they been?	1. Adherence to work plan	1. Project documents – workplan, technical reports
Is ClimBeR still appropriately responding to the needs of the partners?	1. Adoption of project innovations 2. # of successful partnerships executed 3. Joint planning & execution of activities with partners	1. Desk review of technical reports 2. KIs with project staff and partners
What changes have occurred in the operational landscape in Kenya, and what adjustments are required in the ToC in response to the new context?	1. Adhere to/changes in project workplans.	1. Desk review of technical reports 2. KIs with project staff and partners
Objective 2: To demonstrate how ClimBeR impacts are emerging from the inputs and the activities, highlighting the drivers of and barriers to participation and uptake.		
How is ClimBeR initiative being managed and how has this style supported researchers, policymakers, and development practitioners to work <i>collaboratively</i> , embrace <i>systems-thinking</i> , and recognize the <i>diversity of livelihood pathways</i> leading to climate resilience?	1. Partnership agreements developed. 2. Diversity of partners involved. 3. Cross work packages and cross initiatives activities 4. Joint activities - work planning (<i>co-production and validation of workplans</i>)	1. Desk review of technical reports 2. Desk review of workplans and project M&E reports, outcome stories, etc. 3. KIs with implementation team

Are ClimBeR products attractive for private sector investment and how far have they attracted investments in climate finance? By whom and how well were the associated activities implemented?	1. Adoption of ClimBeR innovations/products by private sector 2. Value of private sector investment in ClimBeR innovations	1. KIIs with project staff and partners 2. Desk review of technical reports, outcome stories, etc.
Is ClimBeR helping stakeholders to embrace the concept of gender and social equity, and the disruption to balances of power that may be associated with the transformation?	1. Influence of ClimBeR in mainstreaming of gender and social equity consideration in products and processes of partners/stakeholders	1. Desk review of M&E reports and technical reports 2. KIIs
How are the WPs and crosscutting themes linked?	1. Mainstreamed or embedded cross cutting activities in work package work plans 2. Output indicators for cross cutting themes reported under different work packages	1. Desk review of workplans and technical reports 2. KIIs
How is ClimBeR working together with other relevant initiatives?	1. Cross initiatives activities 2. Mapping of activities and locations for synergy	1. Desk review of workplans and technical reports 2. KII

2.3.1 Sample and Sampling Design

The evaluation was largely based on qualitative approaches, mainly FGD and KII, supplemented with secondary information from project documents. FGDs were held with farming communities where RCC was being implemented. The FGDs participants were sampled to ensure representative views of diverse groups including women, youth and people living with disability. In all, three FGDs were held, each having 8-12 participants. Key Informant Interviews were held with project staff, project partners, implementing agencies, representatives of national and sub-national governments. A total of 26 Key Informants were interviewed. The key informants, external to ClimBeR, were identified with the help of ClimBeR teams who had worked closely with them.

For quantitative household survey, mainly meant to determine drivers of and barriers to uptake of the various products, we sampled the farm households in Embu where the bundled products had been rolled out. Following Cochran (1977), the sample size was computed as follows:

$$n = \frac{Z^2 pq}{e^2},$$

Where:

n=Sample size required; P=Estimated proportion of population affected, as a decimal; q=1-p; Z=Z-score at the desired confidence level; and e= margin of error.

Because the proportion of population affected was not easy to determine, we used the conservative figure of 50%. At 95% level of confidence, Z=1.96. Thus, allowing for a margin of error of 5%, the sample size for study was computed below:

$$n = \frac{(1.96)^2(0.5)(0.5)}{0.05^2} = 384 \text{ target respondents}$$

However, the survey realised 355 respondents, a success rate of over 90%.

2.4 Analysis of the drivers of uptake of the bundled services

In the ClimBeR project and from literature, crop insurance and agricultural credit access are seen as being complementary. Insurance reduces the risk of defaulting on agricultural credit. Thus, farmers with crop insurance would, theoretically, be more likely to access agricultural credit related to the insured crops. We, thus, analyze the two services using a bivariate probit model, specified as follows:

$$CI_u^* = \delta_1 F_u + \alpha'_1 X_1 + \varepsilon_c, CI_u = 1 \{if CI_u^* > 0; 0 \text{ otherwise}\} \quad (1)$$

$$F_u^* = \delta_2 CI_u + \alpha'_2 X_2 + \varepsilon_F, F_u = 1 \{F_u^* > 0; 0 \text{ otherwise}\} \quad (2)$$

$$\{\varepsilon_c, \varepsilon_F\} \sim BVN\{(0,0), \sigma_c^2, \sigma_F^2, \rho\}, \quad (3)$$

Where ρ is the correlation, σ_i^2 is the variance, CI_u^* and F_u^* are observed binary (latent) variables indicating household uptake of crop insurance and agricultural finance (credit), respectively, X_1 and X_2 are vectors of explanatory variables while ε_c and ε_F are the error terms for the respective equations.

The reduced form of the model needed for consistent and efficient estimates is as specified below:

$$CI_u = \pi'_1 X + \mu_c, \quad (4)$$

$$F_u = \pi'_2 X + \mu_F, \quad (5)$$

$$\{\mu_c, \mu_F\} \sim BVN\{(0,0), \sigma_c^2, \sigma_F^2, \tau\}, \quad (6)$$

Where τ is the measure of the extent to which the errors covary, σ_i^2 is the variance and X is the union of the exogenous variables in the system. The correlation of the error terms arises when there are unobserved factors which influence uptake of crop insurance as well as agricultural credit. In the presence of such correlation, univariate probit yields biased estimates while bivariate probit leads to efficient and consistent estimates (Greene, 2000). Bivariate probit is estimated through maximum likelihood.

The coefficients of bivariate probit cannot be interpreted directly. Thus, we compute the marginal effects, interpreted as change in predicted probabilities associated with the changes in exogenous variables.

3. Key Findings

3.1 Implementation context

Kenya is heavily dependent on rainfed agriculture, characterised by low and erratic productivity, leaving households food and nutrition insecure. The low productivity has been attributed to low adoption of improved farm technologies, declining soil fertility due to shrinking land sizes and intensive monoculture and the increasing frequency and intensity of weather extremes. While the smallholder farmers have devised climate change coping strategies, their efforts have not yielded transformative changes because they are not based on sufficient scientific information, and more often than not, their resources are insufficient to enable them to respond adequately. In marginal areas, climate change has led to increasing conflicts especially between the pastoralists and the crop farmers, leading to increased insecurity in Counties such as Baringo, West Pokot, Turkana, Samburu, Laikipia, Isiolo, Garissa, Kitui and Taita Taveta. Thus, ClimBeR project is a timely initiative, seeking to ensure systemic transformation for enhanced resilience, productivity, and equity by delivering science and innovation to simultaneously transform food, land, and water systems in a changing climate. The project is building on the milestones of the participatory action research by CGIAR Research Program on Climate Change, Agriculture and Food Security (CCAFS).

The implementation of ClimBeR started in June 2022, a period of heightened political campaigns, culminating in change of political leadership. This may have slowed down activities which required authorisation from the executive and/or dovetailing to government's development agenda. Certain policy and economic shifts have also been witnessed that may have some ramifications on the project:

1. Rapidly and sharply increasing interest rates may undermine interest in RCC products, thus compromising uptake of bundled financial products.
2. Input (fertilizer) subsidies may undermine uptake of RCC as participating input suppliers are not offering subsidized inputs.

3.2 Implementation approach

In implementing the various Work Packages and/or the cross-cutting themes, ClimBeR team uses any or a combination of the following three approaches:

1. Participatory/consultative approach involving different stakeholders e.g., researchers, input suppliers, financial institutions – banks and insurance companies. This is the most widely used approach, mainly in co-creating solutions/innovations such as bundled financial products, voices of change, unpacking iFEED results into policy recommendation, CSA MSP.
2. Top-down approach, used mainly in development of technical products e.g., CSO, iFEED modelling and credit scoring tool which are then socialized among the stakeholders for adoption.
3. Bottom-up, putting downstream actors at the core of product development/delivery e.g., development of climate security training module, use of “champion of change”, integrating climate security into National Climate Change Action Plan (NCCAP) III.

Overall, however, ClimBeR applies an agile management and implementation style with a strong focus on the ToC as the guiding instrument. ToC recognizes complexity of climate change concerns and

multidimensionality of attendant solutions and thus emphasizes the contribution of diverse expertise and capabilities. This has the following effects:

1. *Working collaboratively* – by recognizing complexity of issues and contribution of diverse expertise, ClimBeR encourages interdisciplinary approaches which foster collaborative engagement across disciplines.
2. *Systems thinking* – ToC maps out various pathways to achieving an intended outcome, thus revealing the multi-dimensionality of solutions to complex problems. This underscores that “the whole is greater than the sum of its parts” and that solutions are more impactful if each part is brought to bear on the collective outcome. Thus, solutions are viewed as a system and not a sum of different parts.
3. *Diversity of livelihood pathways* – ToC recognizes that desired livelihood outcomes are realized through multiple pathways based on multidimensionality of problems and the attendant solutions.

Also notice that, by design, the ClimBeR Work Packages do not work in silos. Instead, they communicate with each other and, where appropriate, they collaborate on specified issues. More importantly, the cross-cutting themes are deliberately designed to work with and add value to the Work Packages. This has been manifested in the following:

1. GSE is linked to WPs through mainstreaming of processes (NCCAP III, IGAD Climate Change Adaptation Strategy, etc.) and outputs. This is achieved through the:
 - a. ToC that outlines relevant gender considerations across WPs
 - b. Deliberate extra effort of the GSE implementing team to reach out to the Work Packages and go beyond planned activities.
2. Climate Finance: Climate finance is linked to WP1 and WP2 through the following innovations.
 - a. Digital climate finance scoring tool – which is expected to be used by Banks to assess creditworthiness of client farmers.
 - b. Financial and climate literacy program for training farmers and motivating them to take up RCC.
 - c. Climate finance has also contributed to the development of GCF proposal.
 - d. Related to GCF proposal is the use of the CSAIP which was developed through AICCRA, although it is yet to be signed by the PS of the Ministry of Agriculture and Livestock Development.

Beyond the inter-work package linkages and the cross-cutting themes, ClimBeR also works with other related initiatives. The CGIAR Research Initiative on Livestock and Climate (L&C) is one such initiative which has work with ClimBeR on various issues, particularly to address the double-burden of vulnerability of livestock-based livelihood to climate change and the contribution of livestock to GHG emission. These issues are common to the resilience building focus of ClimBeR and this has motivated the two initiatives to work together. L&C contributes to the development of SE tool for case studies that sought to identify household vulnerabilities to climate change to inform appropriate advocacy that would enhance resilience of respective households to climate change.

3.2.1 Implications of the Implementation Approaches

The implication of the implementation approaches is that ClimBeR must put the partners at the centre of its delivery system and respond effectively to their interests. This has resulted into:

1. Alignment of ClimBeR's activities with the mandate of partner organizations. For example, RCC product extends possibilities to increase client base –RCC has increased farmers appetite for credit; increased clientele base for insurance companies (i.e. smallholders are more aware of and more interested in insurance). Family Bank and Equity Bank have also been enabled to venture into /enhance digital lending space, partly because of enhanced efficiency in assessing creditworthiness of clients and partly because of pooling of the smallholders.
2. Strengthened role of partners. For example:
 - i. Strengthened educative role of Mediae – being able to appropriately package weather information to accompany that with relevant advisory on interventions, attributable to:
 - Leveraging technical expertise to identify appropriate advisory that should accompany weather information.
 - Leveraging technical skills to unpack technical weather information into easily understood form.
 - ii. Strengthened advocacy role of CEMIRIDE – advocacy backed up by rigorous research evidence, thus enhancing credibility among other partners and increasing their influence.
3. Partners embracing gender and social equity in their work. ClimBeR is deliberate in mainstreaming gender consideration in their processes and innovations:
 - i. Family Bank deliberately targeted women groups as emphasized by ClimBeR.
 - ii. Equity Bank – IFPRI ensured that gender was integrated into RCC to enable targeting of women farmers for empowerment.
 - iii. CEMIRIDE—ClimBeR has made them more deliberate about gender consideration; going beyond just ensuring participation to look at various dimensions of gender inclusion.
 - iv. IGAD – ClimBeR contributed to include climate, peace, and security evidence in the IGAD Climate Change Adaptation strategy, with strong considerations of gender and social equity perspectives.
 - v. NCCAP III – Climate Security was introduced for the first time in NCCAP III including identification of climate security action plans with gender considerations in the Adaptation Technical Assessment Report (ATAR) annex of the NCCAP III.
 - vi. AU Climate Security Assessment - East Africa Chapter - done by ClimBeR had GSE considerations mainstreamed.

3.2.2 Monitoring and supervision of activities

The ClimBeR implementation approach brings together many partners, researchers, and stakeholders. While some activities are implemented by the ClimBeR staff, others are implemented by external organisations and consultants. This calls for a robust supervision and monitoring mechanisms to

guarantee timely and quality delivery. It was observed that the project uses several approaches and instruments to achieve this, and these include:

1. Periodic meetings within and across Work Packages and cross-cutting themes
2. Working on common platforms
3. Review project-wide meeting.
4. Periodic reports, under MELIA
5. Issue of contracts and sub-contracts with clear deliverables and basing payments on the deliverables
6. Periodic evaluation studies to determine what is being done right, what needs to be modified, what needs to be dropped and what needs to be scaled.

There was consensus among the stakeholders that these approaches and instruments have been effective in keeping the project on track and ensuring timely delivery of the project targets.

3.3 Progress towards impact

In this section, we review progress towards outcomes and outputs. We first map out the impact pathways for each Work Package and the cross-cutting themes to understand how activities were intended to yield outcomes. We then identify evidence of progress and changes in section 3.3.2 before evaluating the contribution of ClimBeR to observed changes.

3.3.1 Impact pathways by Work Package

Impact pathways are central to process evaluation – mapping out how project *inputs/resources* are used in various *activities* across different intervention areas to deliver *outputs* that could contribute to *outcomes* and eventually to long-term *changes or impacts*. Since ClimBeR is in the middle of its implementation timeline, it is likely that most interventions have not progressed to impacts. However, we assume that activities implemented so far have resulted into outputs that can reveal if the project is on track to achieving outcomes and impacts.

Work Package 1: Reducing risk in production system-linked livelihoods and value chains at scale.

Work Package 1 (DE-RISK) aims to reduce livelihood and value chain risks through a blend of complementary interventions including agricultural risk management, digital agroclimatic information services, and climate smart agricultural innovations. Additional work involved modelling production systems typologies with a nutrition lens to enable prioritization and targeting of investment. Provision of climate information is accompanied by information on relevant climate smart innovation, which triggers demand for relevant innovation. On the other hand, provision of risk-contingent credit strengthens risk appetite and demand for credit that enhances adoption of relevant technologies.

Previous approaches to climate change adaptation such as exposure to climate smart technologies have led to sub-optimal uptake of relevant practices due to lack of appreciation of complementary interventions. Consequently, smallholder production systems lack resilience to severe climate change effects such as floods, drought, and high temperatures with many poor households in Kenya unable to adequately adapt to climate change. There is also limited financing of climate change adaptation with very little if any finances flowing directly to vulnerable households. Moreover, existing financial service products do not adequately integrate the risk mitigating benefits of CSA technologies in assessment of risk levels.

To address the gaps, ClimBeR aimed to develop and avail risk-contingent credit (RCC), develop digital systems for provision of climate information services, and model production systems typologies that enable targeting of investments to gender outcomes. Through these innovations, households are expected to access bundled financial products, and climate information that would enable them to adopt climate-mitigation and adaption practices and therefore enhance their resilience to climate change. See Figure 3 for illustration of impact pathways.

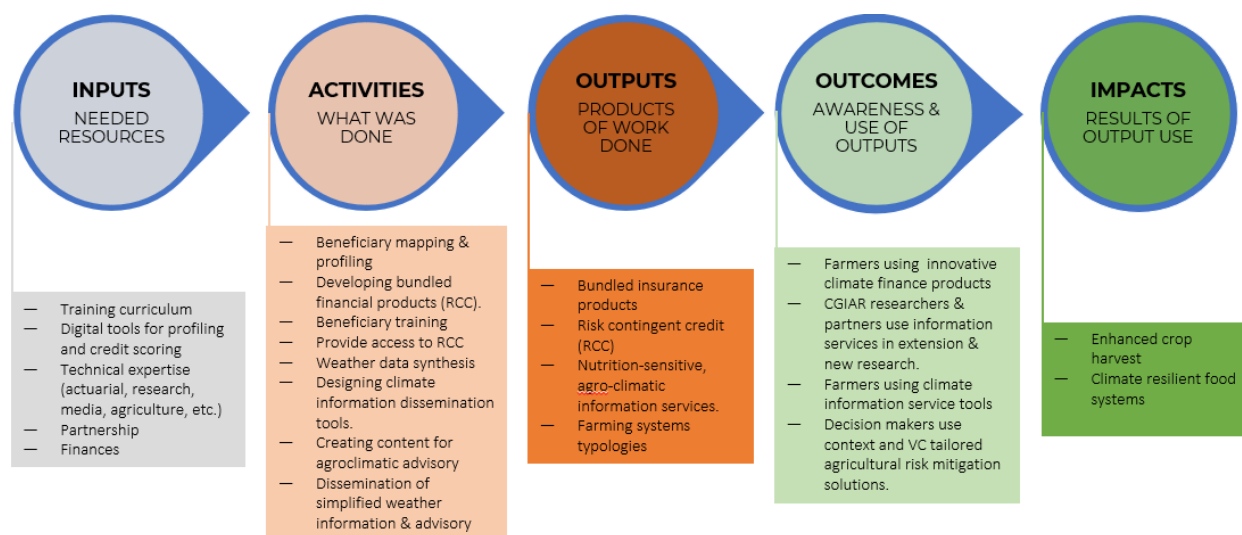


Figure 3. Impact pathways for Work Package 1

ClimBeR has made tremendous progress, working with financial institutions (Family Bank, Equity Bank and APA insurance) to develop and pilot risk contingent credit (RCC) among farmers in Embu and Machakos counties. ClimBeR has also worked with MEDIAE to develop synthesized weather information that is accompanied with advisory on appropriate interventions and practices. The information is disseminated via Shamba Shape Up episodes on television and iShamba platform. Moreover, farmers are also receiving training on financial literacy and agronomic practices, all aimed at enhancing uptake of RCC.

To deliver on this goal, ClimBeR has leveraged diverse expertise including in research, media, and finance or actuarial science. These achievements have also been realized courtesy of partnership with various organizations including banks, media, and academic institutions.

Work Package 2: Building production-system resilience- through recognizing the relationship among climate, agriculture, security, and peace.

Work Package 2 is premised on the recognition that climate change and variability may trigger competition over natural resources thus exacerbating risks to peace in agricultural communities. Climate, peace, and security considerations in relation to agricultural livelihoods therefore aims to foster the resilience of production systems to climate-induced security risks and instability. As such, ClimBeR aims to generate scientific evidence on climate, peace, and security, mainstreaming analyses on the topic into agricultural programming, facilitating policy coherence and integration between climate, security, and agricultural policies, and prompting conflict-sensitive climate financing.

Given the “risk multiplier” effect of climate change and variability, it is imperative that climate adaptation factors in potential risks to peace that may be exacerbated by the ongoing climate crisis.

While the concept of climate, peace, and security is increasingly accepted, efforts to address security concerns in Kenya are still pursued in isolation, neglecting the intertwined linkages between climate change, agricultural livelihoods, and instability. Similarly, adaptation and mitigation measures against worsening climatic shocks are often pursued with limited consideration of security threats.

To address these gaps, ClimBeR developed specific tools for qualifying, quantifying, and monitoring the interlinkages between climate, peace, and security within agricultural settings. The innovations developed should serve as tools for decision-makers and local stakeholders to improve their understanding on the role of climate change as a risk multiplier in agricultural communities. These innovations include a Policy Coherence Analysis Tool, a Climate Security Sensitivity Tool (CSST), the [Climate Security Observatory](#) (CSO), the Climate Security Index (CSI), and Climate Security Programming Dashboard among others. ClimBeR also ensured that the analysis conducted using these tools cascade into different local contexts, through the identification of context-specific climate, peace, and security gender-responsive pathways that integrate climate adaptation and peace benefits. This involves working with the most vulnerable groups in the face of climate change to ensure that their voices are factored into policy processes and interventions. Through “voices of change”, ClimBeR empowers communities to better understand and raise awareness on how the livelihoods of specific vulnerable groups are affected by climate change and why adaptation measures should factor in peace positive contributions.

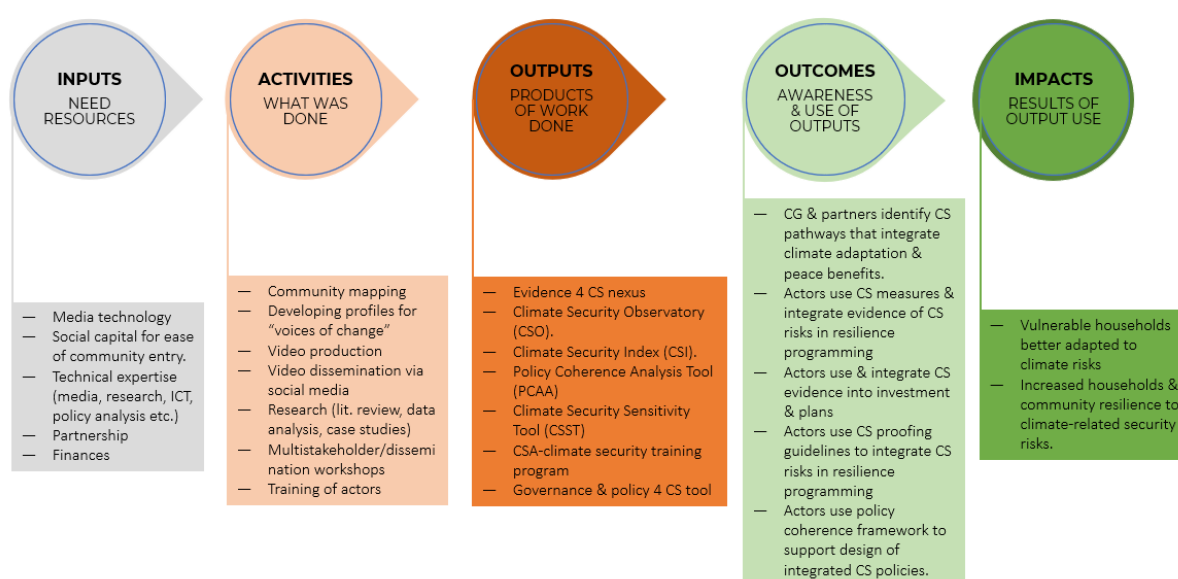


Figure 4. Impact pathways for Work Package 2

These tools, together with on-the-ground validation processes, are also expected to support policy makers in developing coherent and integrated policies to account for multiple and overlapping climate and security risks. At the same time, researchers, funders, governments, and demand partners, leveraging these innovations, can easily integrate climate-related security risks into resilience programming. See Figure 4 for illustration of impact pathways.

ClimBeR has made significant progress in implementing activities along these lines. First, ClimBeR has consolidated a scientific understanding of the interconnection between climate, peace, and security in Kenya as well as identified potential solutions aimed at mitigating the impacts of climate change on risks to peace (Medina *et al.*, 2022). To achieve this, ClimBeR organized a workshop with climate, development, humanitarian, and peace actors, together with governmental representatives, to define

a common vision on climate, peace, and security in the context of Kenya (<https://hdl.handle.net/10568/125809>). This common vision encompasses how climate-related security risks are manifesting across the country as well as which could be potential solutions to the identified threats.

ClimBeR has also adapted the CSO as a decision-support tool. The platform identifies different climate, peace, and security pathways through which climate change can affect intervening socio-economic and political factors, thereby worsening the root causes of conflict. Beyond this theoretical understanding of the nexus, the tool also presents a map for climate-security hotspots – namely areas where climate and conflict risks co-occur and overlap – pointing out the most vulnerable groups. The tool also includes other quantitative analyses on the multifaceted relationships characterizing the climate, peace, and security nexus, and proposes some solutions for addressing the vicious cycle linking climate change and conflict risks (Climate Security Observatory-Kenya; Summary for policy makers).

On the policy side, ClimBeR has developed a *Policy Coherence Analysis Tool* and applied it to assess whether climate, agricultural, and security policies all recognize the role of climate change as a “risk multiplier”. This is more to ensure that policies acknowledge the existence of a nexus between the factors and link them in the process of policy implementation. ClimBeR has also collaborated with the “Alan Turing Institute” to develop Climate Security Index (CSI) – a quantitative framework for measuring and monitoring climate security vulnerability. The developed methodology has been applied to estimate climate security vulnerability of Kenya (Guerrero *et al.*, 2023). ClimBeR has also developed a [Climate Security Sensitivity Tool](#) for ex-ante assessment of climate actions for conflict-sensitivity and peace-responsiveness. The tool empowers practitioners and decision makers who design adaptation initiatives to avoid maladaptation and the related unintended consequences.

Finally, ClimBeR has also worked with the Ministry of Agriculture and Livestock Development, and the Climate-Smart Agriculture Multistakeholder Platform (CSA MSP) to develop a training course titled *Climate, Peace, and Security in the Agriculture Sector: Unpacking the Climate Change, Agriculture, and Peace Nexus*. Preparations are underway to pilot this course in 2024.

Work Package 3: Developing adaptation instruments to inform policy and investments.

Work Package 3 focusses on integrated policy frameworks that combine 3 approaches: a) top-down approaches involving *participatory scenario workshops (in-country task forces and knowledge integration workshops to translate modelling results into policy and investment decisions)*, b) bottom-up processes that *leverages local experts with indigenous knowledge of innovative grassroots practices, and social dynamics to facilitate policy and financing options that support transformation of agrifood systems*. and c) cross-scale climate-smart agricultural water management to evaluate potential trade-offs among climate adaptation options and inform alignment of climate resilience policies with other key development policies.

In Kenya, the focus is on policy pathways for climate resilient, nutrition secure and socially equitable futures. This is premised on the understanding that climate change is likely to exacerbate food and nutrition insecurity. Hence the need for an integrated food systems approach that gives due consideration to climate change concerns. Yet existing integrated assessments of food systems tend to be largely driven by data and modelling processes with limited local information on food and nutrition security. Accordingly, ClimBeR supports a policy process that is informed by a “novel” integrated

assessment framework that combines data and modelling with in-country knowledge and expert academic judgement to explore climate-smart and nutrition-secure food systems scenarios, through a process known as integrated Future Estimator for Emissions and Diets (iFEED).

Unlike existing modelling approaches such as Integrated Assessment Models (IAMs), iFEED uses both top-down and bottom-up approaches and goes beyond the simple assessment of future food production and food security outcomes to quantify future population-level nutrition security. iFEED assessment integrates nutrition security and climate smartness, both of which are neglected in independent climate change impacts and food security scenario studies. By incorporating in-country knowledge and expertise, the approach ensures that developed policies are context-specific. The model work begins with participatory scenario workshops and culminates into *calibrated summary statements summarizing model outputs*, which are then expanded through expert elicitation. Expert elicitation translates the calibrated statements into implication statements with input from local experts.

Ultimately, this process is expected to deliver inclusive policy making processes and equip policy makers to formulate policies that transform food systems in favour of the population segments that are at most risk of climate change. See further illustration in Figure 5.

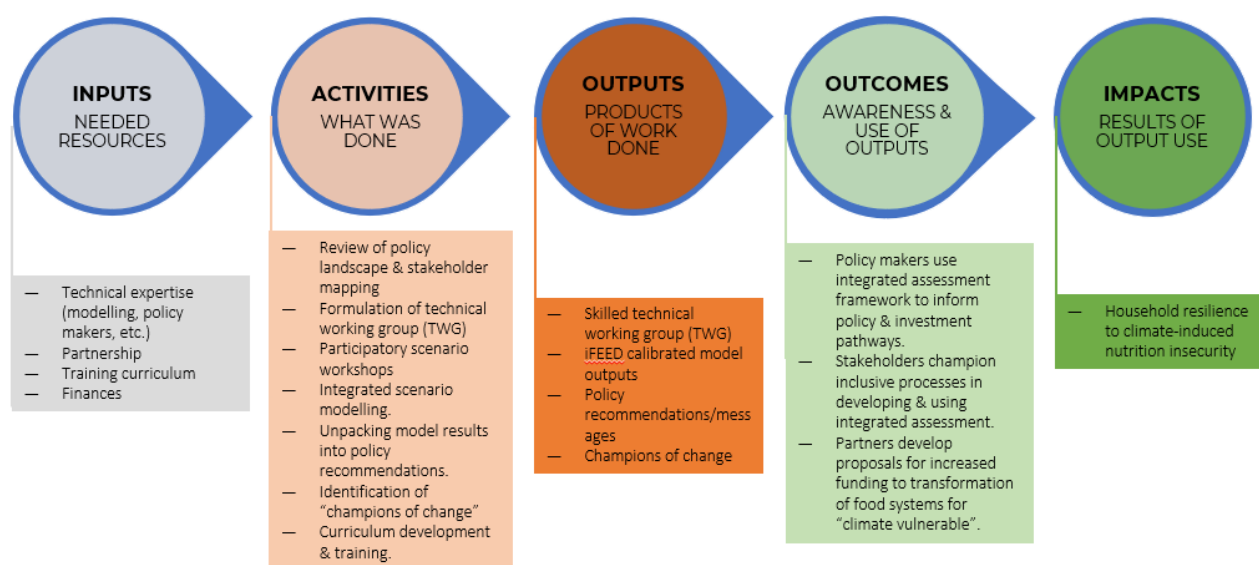


Figure 5. Impact pathways for Work Package 3

Towards iFEED process in Kenya, ClimBeR has already conducted a policy landscape assessment to identify influential actors/stakeholders that should be involved in activities under Work Package 3.

To achieve holistic view of food systems, iFEED leverages diverse expertise, ranging from climate change to food security to develop integrated policies that are transformative, leading to food systems transformation – integrating emission, climate extremes, trade, and nutrition considerations. In Kenya the initial stakeholder engagement brought together research and academic experts from University of Aberdeen, Kenya Institute of Public Policy Research and Analysis (KIPPRA) and experts with knowledge on local food systems including Food, Agriculture and Natural Resources Policy Analysis Network (FANRPAN), in-country agricultural experts drawn from Ministry of Agriculture and Livestock and academic expertise including agro-ecologists and social scientists.

The experts included modellers with knowledge on climate change, food and nutrition and greenhouse gas emissions and local experts with knowledge on local food systems. These experts were drawn from

academic institutions, research centres, food policy organizations and government departments and are accessed via a technical working group (TWG) that is established to facilitate cross-expertise engagement. Local knowledge is leveraged through experts including from academia and in-country agricultural experts. Initial attempt to ride on pre-existing TWG that was affiliated to CSA-MSP was met with challenges hence the recent efforts to form a TWG that is fully dedicated to the policy pathways workstream. The TWG has held 2 meetings so far.

The WP core team are currently discussing the scenarios to explore in nutrition analysis, and this will inform the simulation of future food production for different adaptation scenarios.

Work Package 4: Multiscale governance for transformative adaptation.

Climate change mitigation efforts occur at global scale while adaptation efforts are applied at local scale. Geographically, the contexts within which climate changes occur are also varied and different groups are affected variedly by climate change. Adaptation to climate change therefore calls for governance arrangements that accounts for the variation in contexts and challenges and exploits opportunities for actions in different arenas and varied scales. ClimBeR therefore promotes *multistakeholder polycentric governance (MPG)* structure that enables collaboration across scale and context. ClimBeR is exploring how KIPPRA can be trained on the MPG tool guide and how KIPPRA can adopt some MPG knowledge products for specific policies that are supported by KIPPRA.

To ensure that adaptation measures benefit the most vulnerable, polycentric governance needs to be complemented by adaptation of “*Leave No One Behind*” indicators of SDGs. Capacity to rapidly adjust and apply adaptation measures is further strengthened through ClimBeR’s *Early Warning, Early Action, Early Finance (AWARE) Platform*. AWARE platform goes beyond the traditional early warning systems to translate early warning information into appropriate actions to reduce impacts of disasters. ClimBeR is discussing with National Disaster Management Authority (NDMA) how AWARE can be institutionalized.

Complementary to all this is *Climate Smart Governance (CSG) dashboard* that helps actors to monitor progress of adaptation targets and facilitate exchange of ideas and integration of efforts to achieve transformative adaptation. KIPPRA is engaging National Treasury and National Planning department to explore implementation of CSG dashboard.

The 4 innovations work together to enable relevant institutions to champion integrated frameworks for adaptation, leading to targeted investment in climate smart agriculture by partners and government. The pathway of impact is illustrated in Figure 6.

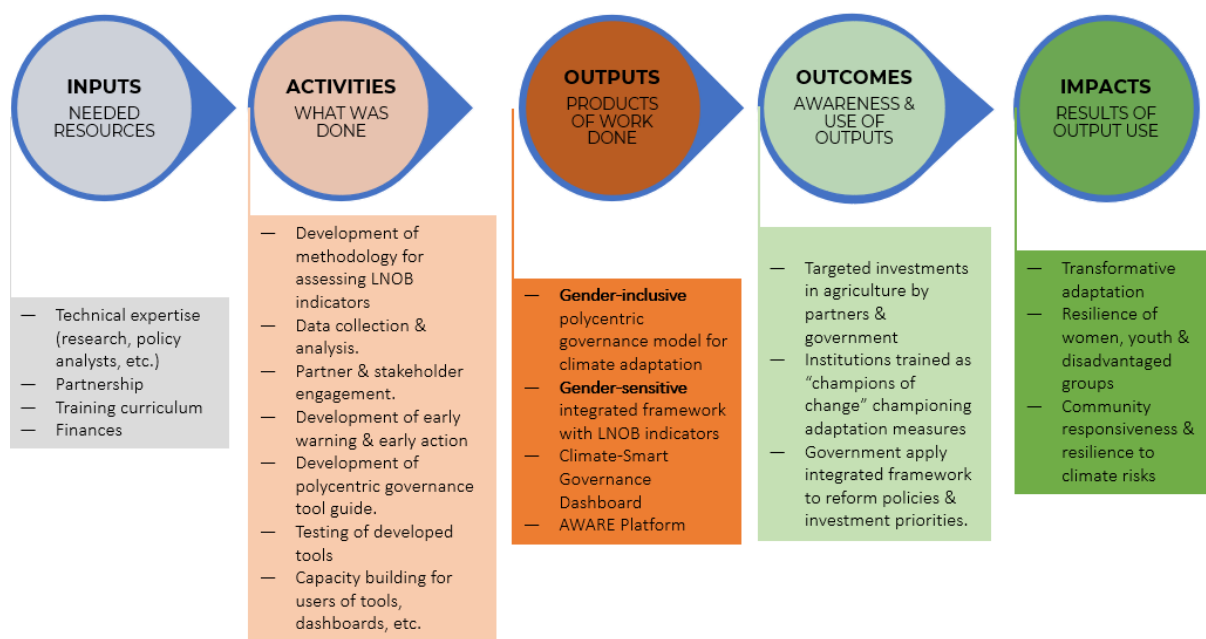


Figure 6. Impact pathways for Work Package 4

Cross cutting themes

ClimBeR is also integrating two cross-cutting themes across the Work Packages – climate finance and gender equity and social inclusion.

Climate finance: Climate finance was intended to be integrated into Work Packages 1, 2, and 3. Work Package 4 did not have direct climate finance-related activities. Towards integration of climate finance across the 3 Work Packages, ClimBeR has developed *climate credit scoring tool* that has been used by the financial institutions to assess creditworthiness of farmers applying for the RCC. The climate finance team also supported the development of *financial literacy training material* that are used by partners to train farmers.

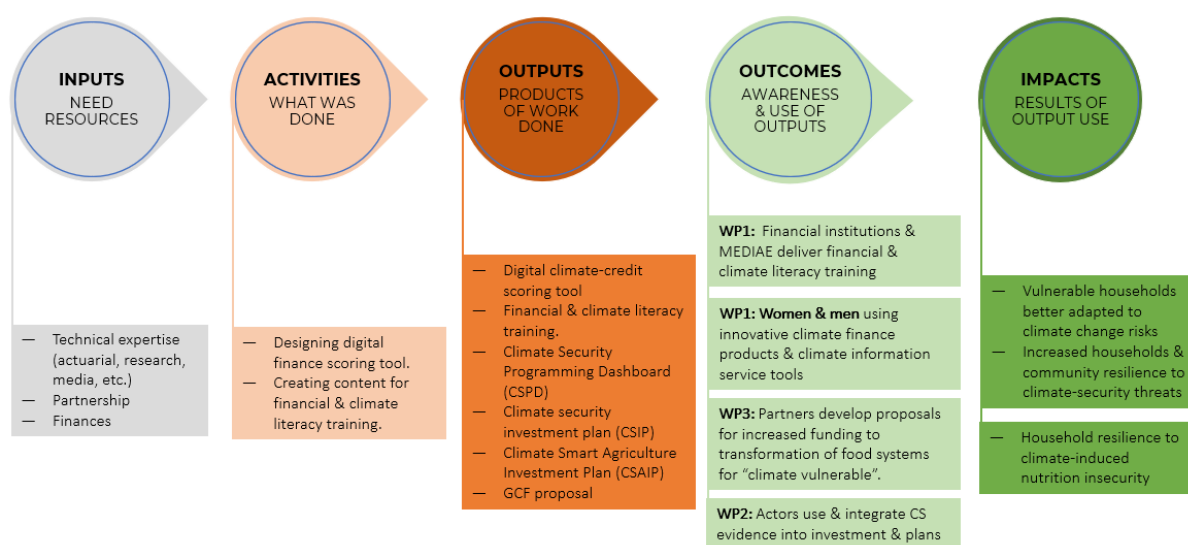


Figure 7. Impact pathways for Climate Finance theme

The team also had substantial input in the financial literacy material that is disseminated via Shamba Shape-Up Program and the iShamba platform. With more financial literate farmers, we expect the uptake of RCC to go up.

Towards conflict-sensitive climate financing, ClimBeR has developed [Climate Security Programming Dashboard](#) that supports executing agencies working with international financial institutions (IFIs) and climate funds to ensure that allocated climate funds reach households that are most vulnerable to “combined effect of climate and security threats”. The tool streamlines risk analyses through conflict-sensitive evaluation that identifies conflict risks and provides tailored strategies for mitigating against these risks.

ClimBeR has also developed *Climate Security Investment Planning* approach that facilitates development of ready-to-finance interventions with dual climate change adaptation and peace dividends. So far, the team has developed a methodology to guide this process across various countries and context (Savelli et al., 2023). This is expected to inform actors to integrate climate security evidence into climate adaptation investment plans all leading to enhanced resilience of households to climate change. See Figure 7 for illustration. Additionally, ClimBeR has also collaborated with the national government through the Ministry of Agriculture, livestock and fisheries to use the *Climate Smart Agriculture Investment Plan for Kenya (CSAIP)* that provides a comprehensive case for investing in agriculture development in the face of climate change for the development of a GCF concept note.

Finally, ClimBeR is strengthening capacity of various county governments to develop proposals that can enhance their access to climate finance. Additionally, ClimBeR has worked with Treasury to localize climate financing and is currently supporting the development of one *GCF concept note for Kenya* covering the counties of Turkana, West Pokot, Narok, Kilifi and Baringo.

Gender equity and social inclusion (GESI): ClimBeR has also been working to ensure that climate change adaptation interventions are inclusive of women, youth, and other groups of interest in communities. The GESI team has therefore developed gender and social equity (GSE) indicators and GSE tools to enable integration of gender concerns in data collection for various studies undertaken by ClimBeR. Training materials were also engendered to ensure better targeting of women with relevant training content.

The tools and innovations developed are expected to increase the proportion of women accessing the bundled financial products while developing gender-sensitive pathways for integrating climate change adaptation with peacebuilding. Figure 8 provides illustration of how the inputs developed and disseminated by GESI team is expected to deliver on specific outcomes.

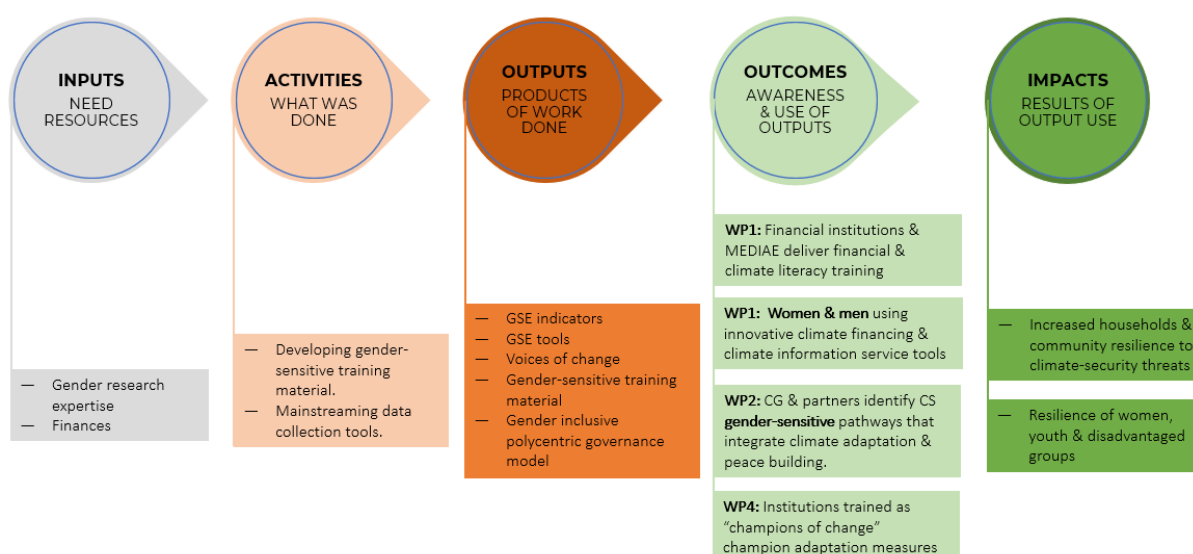


Figure 8. Impact pathways for the Gender and Social Equity theme

The GESI team also facilitated the production of “Voices of Change” aimed at bringing the local knowledge into global decision-making fora. ClimBeR worked with civil society groups to harvest “voices from the field” as part of bringing the voices of interest groups to the decision-making table. See <https://www.cgiar.org/news-events/news/voices-of-change-use-a-rights-based-approach/> and <https://www.cgiar.org/news-events/news/voices-of-change-community-stories-of-climate-adaptation-in-baringo-kenya/>. Working with respective civil society organizations, ClimBeR researchers strengthened and supported the voices of Indigenous People, local communities, and organizations to advocate for climate change adaptation solutions that account for local interests and deliver value to communities that are at most risk of climate change. Notable milestones include participation of these “voices” at COP27 and COP28.

3.3.2 Emerging impacts

Process evaluation involves *identifying significant changes* – outcomes and outputs, *tracing the pathways* to realization of these changes to *build hypotheses on probable causes* of the changes and then subjecting the hypotheses to the 4 tests as outlined in the methodology (Section 2.1). Building on the impact pathways we therefore seek answers to the following three (3) questions:

1. What evidence is there for the intended outcome/outputs?
2. What evidence is there for a contribution to the intended outcome/outputs, if any, by ClimBeR?
3. How significant is this contribution, compared with other possible contributing factors?

In this section, we answer the first question that enables us to establish the emerging evidence of progress.

Work Package 1: All outputs towards derisking smallholder farmers have been delivered as planned except for the development of “*production systems typologies with nutrition lens*”, which requires a rich panel dataset that is not readily available for Kenya. In Ethiopia, Nigeria, and Tanzania where this has been undertaken, the work used Living Standards Measurement Study (LSMS) databases that are not available for Kenya. Nevertheless, the evidence generated from other similar contexts may inform innovations and learning for Kenya.

It is worth noting that some of the outputs from Work Package 1 are already in use leading to some early outcomes. The following evidence show that some of the intended outcomes are being realized, albeit on a lower scale.

1. *Farmers using bundled financial products (RCC)*. Family Bank indicated that out of 3,035 farmers that had been trained, 550 qualified for credit and 260 of them are actively accessing RCC.
2. *Farmers using climate information service tools*. The key tools that were developed and are being used by farmers include “Shamba Shape-Up” and iShamba. Interviews with MEDIAE show that episodes of Shamba Shape-Up are already being aired and iShamba weather messages are being disseminated.
3. *CGIAR researchers and partners use information services in extension & new research*. MEDIAE is already using the developed information services to deliver advisory services to farmers. This is achieved through the Shamba Shape-Up series and iShamba platforms.

Work Package 2: Efforts towards climate, peace, and security have also delivered significant milestones, including the development of planned outputs and application of the same to assess the climate, peace, and security situation in Kenya. Some of the outputs have influenced outcomes – informing the integration of climate security concerns into national and regional programs and strategies. See below for highlights of some of the achievements.

1. *Actors identify pathways that integrate climate adaptation & peace positive contributions*. Towards integrating climate, peace, and security in programs aimed at strengthening abilities of vulnerable communities to overcome threats of climate change and variability, ClimBeR has supported the following activities:
 - i. *Facilitating a common understanding of the climate, peace, and security nexus*. ClimBeR worked with stakeholders to develop a common understanding of the climate, peace, and security dynamics. See “Towards a common vision of climate security in Kenya (Medina *et al.*, 2022).
 - ii. *Capacity building on climate, peace, and security*. A training course on “*Climate, peace, and security in the agriculture sector: Unpacking the Climate Change, Agriculture, and Peace Nexus.*”, has been developed and validated in collaboration with various stakeholders (<https://alliancebioiversityciat.org/stories/resilience-agricultural-communities-integrating-climate-peace-security>).
2. *Integration of climate security into resilience programming*. ClimBeR has also worked with regional and international organizations to integrate climate, peace, and security into programs, and strategies for resilience. These achievements are captured in the documents below:
 - i. Climate Security Index (CSI) to facilitate monitoring the role of climate change as a threat multiplier. See Guerrero *et al.* (2023).
 - ii. The IGAD Climate Adaptation Strategy (2023-2030) was prepared with substantial research inputs from CGIAR and other partners. ClimBeR worked with IGAD to co-develop the Climate security factsheet for Kenya (Factsheet 2023/1). The document applies climate security pathways analyses to outline pathways through which climate exacerbates root causes of conflict in Kenya, focussing on the different hotspots- the Laikipia conservatory and larger northern Kenya. See: https://igadcaep.org/wp-content/uploads/2024/02/FS_IP_KENYA_070723.pdf

- iii. *African Climate Security Risk Assessment*, which analyses the interlinkages between climate, peace, and security across the African continent with a view to informing interventions to address climate-related security risks, looking at human security and sustainable peace. See: <https://climate-diplomacy.org/magazine/conflict/african-climate-security-risk-assessment-executive-summary>
3. *Policy coherence framework*. Towards coherent and integrated climate, peace, and security policies, the following have been realized:
- i. Development of “Climate Security Policy Coherence Toolkits”, which have been applied to assess policies and strategy documents from climate-, peace- and security-related sectors in Kenya.
 - ii. ClimBeR also contributed to include climate security actions in the “National Climate Change Action Plan (NCCAP) III”.

Work Package 3: Implementation of activities under Work Package 3 has progressed at a comparatively slow pace. The slow pace of implementation has been occasioned by delay in identifying and onboarding a local partner to drive the process. There is, however, some progress with on-boarding of KIPPRA and recent establishment of a Technical Working Group (TWG) to steer the process. A few more preliminary activities have also taken place, including:

1. Workshop to identify nutrition indicators for the iFEED modelling.
2. Leveraged lessons from previous iFEED processes in Malawi, South Africa, Tanzania, and Zambia to develop a comprehensive, cross-scale and equity-sensitive picture of nutrition secure pathways in Kenya.
3. Two (2) recent meetings by the TWG. It is imperative to note that the TWG activities were affected by the delay in the formation of the current TWG after initial attempts to work with a pre-existing group failed due to the need for an expanded membership of the Technical Working Group.
4. Busia County: A workshop has been organised with departments in the Busia Municipal Council (Busia County) to implement iFEED modelling framework at the local level. The outcome of the workshop includes identification of policy and county investment initiatives for iFEED result inputs. Stakeholders also suggested strategies of packaging iFEED evidence for specific policies and investment opportunities that were identified.
5. Identification of scenarios for nutrition analysis.

Other than the TWG, these activities are yet to deliver the expected outputs.

Work Package 4: Among the 4 outputs that were intended under Work Package 4, substantial progress has been made with respect to two outputs- development of “Leave No One Behind” (LNOB) indicators for ClimBeR initiative and polycentric governance model. Contacts have recently been made with National Drought Management Authority (NDMA) as a potential partner for the establishment of the *AWARE platform*, an activity that is planned for 2024. With regards to Climate-Smart Governance Dashboard a technical guide has recently been developed in December 2023 (Amarnath, Alahacoon, *et al.*, 2023). While the dashboard is expected to be implemented in all ClimBeR focus countries, implementation has only happened in Senegal, Zambia, and Sri Lanka (Amarnath, Alahacoon, *et al.*, 2023).

1. *Leave No One Behind (LNOB)*. Towards LNOB, ClimBeR has assessed the requirements for disaggregated data that can be used in measuring progress on these indicators. The focus of the

assessment was to explore availability of such data from various sources, including the UN and national statistical systems. A report of the assessment exercise recommends developing project based LNOB indicators that are also aligned with the SDGs. Consequently, a set of 12 such ClimBeR project-based indicators have been developed and disaggregated data on these indicators can be generated to meet the requirement of ClimBeR initiative (Chakrabarti et al., 2023).

2. *Polycentric governance model.* With regards to polycentric governance model, ClimBeR launched the **LocaAlly led Climate adaptation ChampION (ACTION)** initiative and Water Integrity Network (WIN) in Kajiado, Kenya to assess how local communities can take ownership of adaptation strategies through the concept of polycentric governance. The study involved a local water project in Kenya, which explored how the solution was identified, its funding and its implementation all with the aim of understanding involvement of the local community in the process (van Koppen *et al.*, 2023). The assessment underscores the contribution of *inclusive community participation* to ensuring that climate adaptation funding from large global organizations reaches those most at risk.

Climate finance: ClimBeR has made substantial progress, integrating climate finance across the intended Work Packages. All outputs have been delivered as planned with some already being used to influence various actors.

1. *Digital Climate-Credit Scoring Tool.* The tool integrates climate modules into the traditional credit assessment to provide a holistic understanding of a farmer's creditworthiness and ensure accurate evaluation for lending decision in the face of climate risks. See: <https://alliancebioversityciat.org/stories/climate-credit-scoring-tool-revolutionizing-agricultural-finance-africa> and Vyas et al. (2023) for application to dairy farmers in Kenya.
2. *Financial literacy training material.* ClimBeR has produced several training materials on financial literacy that are used by partners to train farmers.
3. *Climate Security Programming Dashboard for Climate Finance (CSPDxCF).* This is an all-in-one solution for assessing conflict-sensitivity of climate mitigation and adaptation projects to ensure that such projects are designed to increase the potential for contributing to peace. See <https://cspd.cso.cgiar.org/intro>.
4. *Climate Security Investment Plan (CSIP).* This an approach for developing ready-to-finance climate action interventions that have a dual objective of climate change adaptation and peace benefits. See: <https://hdl.handle.net/10568/127600>.
5. *Climate Smart Agriculture Investment Plan.* Provides a comprehensive case for investing in agricultural development in the face of climate change and variability. Developed through a stakeholder participatory process, CSAIP documents scientifically screened investment portfolio aimed at tapping into financial opportunities for private sector actors, public institutions and international donors who aim to transform the agriculture sector. See: <https://hdl.handle.net/10568/128309>.
6. *Green Climate Fund (GCF) proposals.* In support of GCF proposals, ClimBeR has been strengthening the capacity of county governments to develop proposals that can enable them to access climate finance- <https://hdl.handle.net/10568/136129>. To further unlock access to climate financing from multilateral agencies, ClimBeR has been working with national (Treasury) and county governments to localize climate change financing through establishment of County Climate Change Funds (CCCFs) that enable mainstreaming of climate change adaptation into local planning and budgeting. This is further supported by National Treasury that provide finance for climate actions and risk

management capacities of County Governments and [The Financing Locally Led Climate Action \(FLLoCA\) program](#).

Gender Equity and Social Inclusion: To ensure that climate change adaptation interventions are inclusive of interest of diverse groups in communities, ClimBeR has facilitated activities aimed at amplifying voices of these groups of interest. ClimBeR also worked with various stakeholders to frame issues of social equity in building resilience to climate variability and extremes.

1. *Voices of change* to “amplify local voices” and strengthen their participation in design of climate change adaptation interventions. Notably, some of these “voices” were facilitated to participate at COP28 (<https://www.cgiar.org/news-events/news/ClimBeR-at-cop28-ensuring-that-local-voices-matter/>).
2. *Social equity in building resilience to climate variability and extremes*. ClimBeR has worked with various stakeholders to mainstream gender and social equity across the various workstreams. This has been achieved through framing of social equity issues that are of concern to climate resilience building. See <https://hdl.handle.net/10568/125398>.
3. *Gender and social equity indicators*. To enable assessment of progress in integration of gender concerns in climate adaptation measures, ClimBeR has developed GSE indicators to guide the development of tools for data collection. See <https://hdl.handle.net/10568/128662>.

3.3.3 Process impact (process tracing)

In this section, we demonstrate the contribution of ClimBeR to the changes or progress that has been realized and assess the significance of such contribution through a series of tests as outlined in the methodology section. We undertake this exercise by Work Package.

Work Package 1: Key achievements include development and deployment of bundled finance products – risk contingent credit; development and deployment of climate information service tools; and use of the developed tools and services by researchers and partners. For each of these, we provide some testimonials below to demonstrate the contribution of ClimBeR.

1. *Bundled financial products - RCC*: Interviews with Family bank indicated that, “*Collaboration with ClimBeR enabled de-risking of farmers through embedded insurance product, which made smallholders attractive borrowers*”. Additionally, “*ClimBeR provided credit guarantee, which gave Family Bank the confidence to lend ...*”. Moreover, Family Bank also emphasized that they would not have achieved RCC without support from ClimBeR since banks do not have the luxury to test products that carry a lot of uncertainties. “*We often look at return on investment and may not want to make small efforts that target just a few potential clients*”.
2. *Climate information services*: The information services were developed in collaboration with MEDIAE, who indicate that collaboration with ClimBeR has enable them to “*unpack technical weather data into simpler messages that are easily internalized by their audience*”. Additionally, MEDIAE indicated that “*they leveraged research partnership to identify appropriate advisory and technical intervention that accompany their weather advisory*”.
3. *CGIAR researchers and partners use information services*. ClimBeR researchers are providing the relevant advisory, especially on interventions, that accompany the iShamba weather messaging provided by MEDIAE. ClimBeR also reviews and validates the technical details of the Shamba Shape-Up episodes. Finally, ClimBeR contributes to unpacking technical weather data received from the meteorology department into a format that is easily understood by farmers.

Towards *evaluating the significance of ClimBeR's contribution* to the changes above, we subject the evidence to the 4 tests as described in the methods (Table 3). For the bundled financial products, we hypothesise that ClimBeR influenced the deployment of RCC by both Family and Equity banks. Apart from reports and academic documents that demonstrate long term research by CGIAR on RCC (Shee *et al.*, 2019), interviews with Family and Equity Bank confirmed contribution of ClimBeR. Indeed, Family Bank emphasized, “... we would not have achieved RCC without support of ClimBeR. Banks do not have luxury to test products that could be characterized by uncertainties.” This evidence of greater influence of ClimBeR on rollout of RCCs passes “*double decisive*” test and therefore confirms the main hypothesis.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw- in-the- wind	Hoop	Smoking -gun	Double decisive
Development and deployment of RCC for use by farmers.	Main hypothesis: ClimBeR influenced the development and deployment of bundled financial products by Equity and Family banks.				
	Clue: ClimBeR researchers have been working on RCC; Family Bank testified that they would not have rolled out RCC without support from ClimBeR.				
	Conclusion: evidence passes the <i>double decisive</i> test, therefore confirms the hypothesis.				
	Rival hypothesis: Equity and Family banks were mainly responsible for the deployment of the bundled financial products that they worked on with ClimBeR.				
	Clue: See links to respective banks' insurance programs.				
	Conclusion: This is a <i>straw-in-the wind</i> clue that does not confirm the rival hypothesis.				

The rival hypothesis is that Family and Equity Banks were largely responsible for the rollout of RCC and that ClimBeR played a less influential role. Evidence shows that both banks have crop insurance products (<https://equitygroupholdings.com/ke/insure/crop-insurance> - Equity Bank and <https://crm.familybank.co.ke/Products/Details/1402> for Family Bank). While these could easily have been developed into RCC, there is no evidence of such efforts. Moreover, these products remain listed in respective banks' website as standalone products. The inferred interest in risk management is a **necessary** precondition for investment into a risk management solution for client farmers but is **insufficient** evidence that it led to action – development of RCC. It is therefore a **necessary but insufficient** evidence that only passes the *hoop* test but does not confirm the rival hypothesis. Therefore, while there is evident intention of the banks to address risk among clients through provision of insurance, it is their collaboration with ClimBeR that was largely responsible for the rollout of the RCC.

With respect to deployment of climate information services, the main hypothesis is that ClimBeR had greater influence in development and dissemination of agroclimatic information by MEDIAE. While MEDIAE has existing platforms – Shamba Shape Up (SSU) and iShamba with running episodes and weather messaging, MEDIAE acknowledges the role of ClimBeR in sharpening their agroclimatic weather information services: <https://hdl.handle.net/10568/126423>. Sophie of MEDIAE said, “We leveraged research expertise from ClimBeR to unpack complex weather data into informative weather

messages and also leveraged technical expertise to identify relevant interventions for agro advisory that accompany weather information.” While MEDIAE had the platforms - SSU and iShamba, ClimBeR played a major role in developing the reliable agroclimatic information. This was a necessary step in dissemination of the information – without this information, then the platforms would not be useful. However, the information alone **is insufficient**; you need the platform to deliver the information. So, this evidence passes the *hoop* test and affirms the relevance of ClimBeR in delivery of agroclimatic information but **does not confirm the main hypothesis**.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw- in-the- wind	Hoop	Smoking -gun	Double decisive
Development and deployment of climate information service tools for dissemination of “informative” agroclimatic information to farmers.	Main hypothesis: ClimBeR was solely responsible for the development and dissemination of agroclimatic information by MEDIAE.				
	Clue: Improved agroclimatic weather information services produced with support from CGIAR.				
	Conclusion: This testimonial passes the <i>hoop</i> test and confirms the main hypothesis.				
	Rival hypothesis: MEDIAE was largely responsible for production & dissemination of agroclimatic information.				
	Clue: MEDIAE has a traditional focus on climate change adaptation that they built on to develop and deploy agroclimatic information services.				
	Conclusion: This could have inspired the work on agroclimatic services – passes <i>hoop</i> test but does not confirm the hypothesis.				

The rival hypothesis is that MEDIAE was largely responsible for the production and dissemination of agroclimatic information, with limited involvement if any of ClimBeR. In any case, climate change and adaptation has been a core focus of MEDIAE. See <https://mailchi.mp/9e2401357cf8/mediae-q3-newsletter-16604083>: “Climate Change adaptation methods will continue to be a core theme for the majority of our productions – particularly *Shamba Shape Up*.” Information from iShamba’s website (<https://ishamba.com/#services>) also indicate that “....Seasonal county forecasts are sourced from *Kenya met*”. However, this clue only passes the *hoop* test - the climate change focus is a **necessary** condition for investment in provision of agroclimatic information, but it is **not sufficient** evidence that MEDIAE worked on the innovation with less involvement of ClimBeR. There are therefore strong indications that both ClimBeR and MEDIAE were equally relevant in development and dissemination of agroclimatic services – leveraging each other’s competencies and expertise to jointly deliver the agroclimatic information services.

Work Package 2: Several milestones have been realized with substantial input from ClimBeR. We provide several documentations and testimonials below, which are a proof of contribution of ClimBeR.

1. *Pathways for integration of climate change, agriculture, and peace benefits:* ClimBeR coordinated the development of a course towards integrating Climate-Smart Agriculture (CSA) with peace

efforts. The training course, “*Climate, Peace, and Security in the Agriculture Sector: Unpacking the Climate Change, Agriculture, and Peace Nexus.*” was developed under the leadership of ClimBeR. A workshop to validate the course content was also organized by ClimBeR. See: <https://alliancebioversityciat.org/stories/resilience-agricultural-communities-integrating-climate-peace-security>.

2. *Policy coherence and integration:* Several policies and strategy documents from climate-, peace- and security-related sectors in Kenya have been reviewed for coherence, involving several ClimBeR researchers (Schapendonk *et al.*, 2023).
3. *Integration of climate, peace, and security into resilience programming:* Contribution of ClimBeR in this regard is demonstrated via the involvement of the initiative in the following areas.
 - Climate Security Index (CSI) was produced with leadership of ClimBeR as documented in Guerrero *et al.* (2023).
 - The IGAD Climate Adaptation Strategy acknowledges the contribution of CGIAR to the climate security and environment chapter of the strategy (<https://igad.int/wp-content/uploads/2023/12/IGAD-Climate-Adaptation-Strategy-2023-2030-.pdf>). Reference is made of how CGIAR research has demonstrated the limitation of existing policy documents in translating the appreciation of the climate, peace, and security nexus into actionable policy measures. Abdimajid Nanow of IGAD, underscored that “*the IGAD Climate Adaptation Strategy (2023-2030) was prepared with substantial inputs and feedback from the CGIAR and other partners. CGIAR’s main contribution was on climate security and environment component.*”
 - ClimBeR also contributed substantially to development of NCCAP III, which is acknowledged as having integrated climate security unlike the previous NCCAP I and II. See report by Kenduiywo *et al.* (2023).
 - Moreover, ClimBeR researchers are also heavily acknowledged as co-authors of the African Climate Security Risk Assessment, indicating the influential role that ClimBeR has played in integration of climate security aspects in recent planning efforts. ([https://weatheringrisk.org/sites/default/files/document/Africa Climate Security Risk Assessment Executive Summary 0.pdf](https://weatheringrisk.org/sites/default/files/document/Africa%20Climate%20Security%20Risk%20Assessment%20Executive%20Summary%200.pdf)).

The **significance of these contributions by ClimBeR** is assessed below.

Towards *climate security gender-sensitive strategies and pathways for integrating climate adaptation with peace initiatives in programming approaches*, our hypothesis is that ClimBeR had great influence in development and deployment of the training course. ClimBeR has a focus on both climate security risks and climate adaptation measures based on Climate-Smart Agriculture practices. The desire to achieve greater impact on resilience must have motivated them to explore the synergies between CSA and climate security. This blog <https://alliancebioversityciat.org/stories/resilience-agricultural-communities-integrating-climate-peace-security> also confirms that this work was largely a pursuit of ClimBeR, with other partners brought on board to accelerate delivery. This evidence is *double decisive* and therefore confirms the main hypothesis of ClimBeR’s influential role in developing the training course.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw-in-the-wind	Hoop	Smoking-gun	Double decisive

CS gender-sensitive strategies and pathways that integrate climate adaptation & peace building in programming approaches	Main hypothesis: ClimBeR is responsible for development of the course on “ <i>Climate, Peace, and Security in the Agriculture Sector: Unpacking the Climate Change, Agriculture, and Peace Nexus</i> ”.				
	Clue: ClimBeR already has interest in climate security and Climate-Smart Agriculture as adaptation measures.				
	Conclusion: The desire to achieve resilience inspired ClimBeR to explore a synergy between CSA and CS.				
	Rival hypothesis: Integration of peace efforts, acclimate and agriculture was driven by ClimBeR partners and government.				
	Clue: Focus on CSA may help them appreciate climate security CSA nexus.				
	Conclusion: These partners are largely focused on CSA. Could not have thought how to integrate climate security (not their focus) with CSA.				

The larger focus of CSA MSP and JKUAT on CSA is *straw-in-the-wind* evidence that cannot confirm the rival hypothesis. Moreover, these partners have demonstrated limited focus on climate security and could not have appreciated the synergy between climate security and CSA. The main hypothesis of greater influence of ClimBeR in development of the training course is therefore confirmed.

For *climate security policy coherence toolkit*, we hypothesize that ClimBeR played an influential role in the development of the toolkit. Evidence shows that the framework was developed and assessment for Kenya done by ClimBeR team (Schapendonk *et al.*, 2023). This evidence of sole involvement of the core ClimBeR team in development of the guide and application for the assessment of Kenya policy and strategy documents, passes the *double decisive* test and confirms the hypothesis.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw-in-the-wind	Hoop	Smoking-gun	Double decisive
Climate security policy coherence toolkit	Main hypothesis: ClimBeR played an influential role in development of a “CS policy coherence toolkit”.				
	Clue: Framework developed, and Kenya assessment done by ClimBeR.				
	Conclusion: Publication shows that ClimBeR team solely developed the toolkit and applied it to assess Kenyan case – <i>double decisive</i> !				
	Rival hypothesis: Other organizations working on climate change policy coherence developed and applied the framework.				
	Evidence: Many organizations work on policy coherence and various aspects of climate change – could have inspired ClimBeR.				
	Conclusion: No clear link between their policy coherence work with what ClimBeR has done.				

The rival hypothesis is that other organizations that work on various aspects of climate change policy coherence played a role in the Kenya Climate Security Policy Coherence and Awareness Analysis (CS PCAA). Indeed, there is substantial work on climate change policy coherence, e.g., <https://hlpf.un.org/2022/programme/policy-coherence-for-sustainable-development-and-climate-justice> and <https://reliefweb.int/report/world/policy-coherence-between-social-protection-and-climate-action-initial-findings-global-studies-and-projects>. But these work on other aspects of climate change other than climate security. This is therefore straw-in-the-wind evidence that does not support the rival hypothesis.

Towards *integrating climate security into resilience programming*, our hypothesis is that this was achieved with greater influence from ClimBeR. Evidence on published documents confirm this as explained above and interviews with some partners, e.g., IGAD provide further evidence. Interviews with IGAD management confirm that “.....CGIAR’s main contribution was on climate security and environment component.” Similarly, Kenduiywo *et al.* (2023) show that previous NCCAP I & II lacked integration of CS. These are convincing evidence that confirm substantial role of ClimBeR and therefore pass the *double decisive* test and confirm the hypothesis.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw- in-the- wind	Hoop	Smoking -gun	Double decisive
Integrate CS risks in resilience programming	Main hypothesis: ClimBeR played an influential role in integration of CS in resilience programming by IGAD, NCCAP, etc				
	Clue: Published documents acknowledge the role played by CGIAR. Testimonials also confirm.				
	Conclusion: Evidence clearly shows that ClimBeR participated in processes and brought in CS elements that were not there before. This is <i>double decisive</i> evidence.				
	Rival hypothesis: Respective organizations led the integration of CS in resilience programming				
	Clue: Other organizations in Kenya such as IGAD were already working on climate security - IGAD already appreciated the link among climate change, peace, and security.				
	Conclusion: The interest in CS could have stimulated work on integration but does not confirm that this interest led to the integration work.				

The rival hypothesis proposes that respective organizations integrating CS had greater influence in the process. In any case, organizations such as IGAD were already actively working on climate security. See 2022 policy brief: [https://www.icpac.net/documents/644/Policy Brief Clim 2022 final print.pdf](https://www.icpac.net/documents/644/Policy%20Brief%20Clim%202022%20final%20print.pdf). Similarly, Alan Turing Institute that was involved in developing the *Climate Security Index* (CSI) was also working on climate security. See <https://www.turing.ac.uk/research/research-projects/climate-security-index>. However, this work focus alone, is **insufficient evidence** in support of the rival hypothesis. Our main hypothesis of the overriding influence of ClimBeR in realizing this change is therefore confirmed.

Work Package 3: Progress under this Work Package has been slow with only 2 notable preliminary activities – a stakeholder workshop to identify nutrition indicators for iFEED modelling and formation of technical working group. These are, however, very early days for the Work Package and there is therefore no tangible movement towards output that can be subjected to contribution assessment.

Work package 4: This Work Package has worked on 2 out of 4 outputs with varying levels of progress. In both cases, ClimBeR has largely been responsible for the achievements so far.

1. *Leave No One Behind (LNOB)*. Assessment of the SDG LNOB indicators was led by ClimBeR researchers and the report produced by the same team. See Chakrabarti and Amarnath (2023).
2. *Polycentric governance model*. Application of the concept of polycentric governance to a community-scale water tenure in Kajiado, Kenya was led by ClimBeR researchers who also produced the relevant report. See van Koppen *et al.* (2023) and Amarnath, Mweemba, *et al.* (2023).

We assess the **significance of ClimBeR's contribution** to these achievements below following the 4 tests.

With respect to *bottom-up multiscale polycentric governance (MPG) model* for climate adaptation, we hold that ClimBeR played an influential role in conceptualizing and applying the MPG in Kenya. Besides contributing to the publication on applied MPG (Fasting *et al.*, 2021), ClimBeR led the study that applied MPG to uncover the importance of local leadership in the implementation of community projects (Amarnath, Mweemba, *et al.*, 2023). Moreover, ClimBeR implements the ACTION grant program that promotes the MPG model. This is **strong evidence in support** of the main hypothesis – passes the *smoking-gun* test but is **not necessary** criteria for confirming the hypothesis.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw- in-the- wind	Hoop	Smoking -gun	Double decisive
Bottom-up multiscale polycentric governance model for climate adaptation	Main hypothesis: ClimBeR played an influential role in conceptualizing & application of MPG model in Kenya.				
	Clue: ClimBeR played a central role in conceptualizing MPG approach for climate change & agriculture.				
	Conclusion: Deliberate and consistent effort; strong evidence of substantial contribution to achievement.				
	Rival hypothesis: Other organizations influenced the development and application of MPG model				
	Clue: WIN was a collaborator in the study and had provided training to improve governance of local water systems.				
	Conclusion: Interest in locally led governance most likely influenced their contribution. No evidence of their overriding influence in MPG development exists.				

The rival hypothesis states that other organizations working with ClimBeR were responsible for the development and application of the MPG model. While their interest in community led project

implementation could have inspired their contribution to the study, there is **insufficient evidence** that this gave them greater stake in development and assessment of MPG in Kenya. We conclude that there is strong evidence that ClimBeR played an influential role in the development and application of MPG in Kenya.

We finally look at the development of “Leave No One Behind” (LNOB) indicators for ClimBeR. The main hypothesis is that ClimBeR influenced the development of LNOB indicators for ClimBeR. We note that the review of SDG LNOB against the expectation of ClimBeR initiative was exclusively led by ClimBeR. See Chakrabarti and Amarnath (2023). This was followed by development of LNOB indicators for ClimBeR upon realization of the dataset limitation for the initiative countries. This *double decisive* evidence confirms the main hypothesis.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw- in-the- wind	Hoop	Smoking -gun	Double decisive
Leave no one behind	Main hypothesis: ClimBeR has greater influence in development of LNOB indicators.				
	Clue: ClimBeR recognizes the need for indicators and initiated development of LNOB indicators for the project.				
	Conclusion: Evidence overwhelmingly shows exclusive work by ClimBeR on the development of indicators.				
	Rival hypothesis: Other organizations have led development of LNOB indicators for ClimBeR.				
	Clue: Expressed interest on the need for such indicators could inspire the drive towards development of such.				
	Conclusion: Effort stopped at providing guidelines. Cannot lay claim to actual development of indicators.				

The rival hypothesis attributes the development of LNOB indicators for ClimBeR. This could be motivated by their recognition of the need for LNOB indicators. See: https://www.poverty-inequality.com/wp-content/uploads/2021/02/GIZ-2021-Indicators-Matter-to-LNOB_EN.pdf. But even with this recognition, some of these organizations only went as far as developing guidelines for developing such indicators. See: <https://www.unicef.org/media/102136/file/LNOB-in-WASH-Guidance-Note.pdf>. This *straw-in-the-wind* evidence does not confirm the rival hypothesis. We therefore confirm that ClimBeR was overwhelmingly responsible for development of its LNOB indicators.

Climate finance: The cross-cutting work in Finance is mostly evident in Work Package 1 through *Digital Climate-Credit Scoring Tool* that is used to assess credit worthiness of borrowers and *Financial and Climate Literacy Training Materials*. Climate finance has also been embedded in Work Package 2 through *Climate Security Programming Dashboard (CSPD)* and *Climate Security Investment Plan (CSIP)*. Additionally, ClimBeR has strengthened capacity of county governments to access climate financing. We highlight below the contribution of ClimBeR to these achievements.

1. *Digital Climate-Credit Scoring Tool*. ClimBeR worked with Financial Access Consulting Services (FACS) to develop a digital climate scoring tool by embedding climate modules into the digital credit service tool (LendXS) that had been conceptualized by FACS. According to FACS, the partnership leveraged climate data and expertise from ClimBeR's team, and collected additional household/farm level data, using tools that were developed jointly with ClimBeR.
2. *Financial literacy training material*. While the banks have their own financial literacy training material that they usually use, ClimBeR developed additional material that had emphasis on climate information and the associated risks. This was meant to stimulate interest in the bundled financial services.
3. *Climate Security Programming Dashboard for Climate Finance (CSPDxCF)*. ClimBeR has also developed CSPD to guide climate-sensitivity assessment of climate adaptation and mitigation projects. See: <https://cspd.cso.cgiar.org/intro>. Existing evidence shows that the platform was developed by ClimBeR in partnership with Interpeace - <https://reliefweb.int/report/world/cgiar-and-interpeace-launch-innovative-dashboard-promote-peace-through-climate-finance>.
4. *Climate Security Investment Plan (CSIP)*. This approach was developed by ClimBeR to guide development of ready-to-finance interventions that have dual purpose of climate change adaptation and peace benefits. The authorship of the document (<https://hdl.handle.net/10568/127600>) show that all the authors are ClimBeR researchers.
5. *Climate Smart Agriculture Investment Plan*. The plan that provides a comprehensive case for investing in agriculture development in the face of climate change and variability, was developed in partnership with CSA-MSP. The acknowledgement of the document (<https://hdl.handle.net/10568/128309>) reads, "This report, 'Climate Smart Agriculture Investment Plan (CSAIP) for Kenya', was prepared under the joint leadership of CSA-MSP and the AICCRA project with technical support from the Alliance-Bioversity-CIAT".
6. *Green Climate Fund (GCF) proposal*. While support towards development of GCF proposal for Kenya that targets 5 counties is still going on, ClimBeR is running a capacity building program for counties, strengthening capacity to unlock climate finance. See <https://hdl.handle.net/10568/136129>. ClimBeR also works in collaboration with National treasury to localize climate change financing.

We assess below the **significance of the contributions by ClimBeR** to some of these achievements.

For *digital climate-credit scoring tool*, we hypothesize that ClimBeR played an influential role in development of this innovation. According to FACS, the previous tool they had – LendXS, did not have the climate modules. FACS therefore leveraged climate data and expertise from ClimBeR's team, and collected additional household/farmer level data, to enhance the tool. Simeon Brouwer of FACS confirms that "Although we already had the product idea even before going into partnership with ClimBeR, we could not have developed the module without their data, expertise and networks". This testimony is double decisive evidence in favour of the hypothesis that ClimBeR had overriding influence in the development of the innovation and therefore confirms the hypothesis.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw-in-the-wind	Hoop	Smoking-gun	Double decisive
Digital climate-credit scoring tool	Main hypothesis: ClimBeR played an influential role in development of <i>digital climate-credit scoring tool</i> .				

Clue: Partners indicate that they “would not have developed the module without the data, expertise, and networks” brought about by the partnership with ClimBeR. Conclusion: Testimony is double decisive and therefore confirm the hypothesis.				
Rival hypothesis: FACS was mostly responsible for development of <i>digital climate-credit scoring tool</i> .				
Clue: FACS admit that they already had the product idea but goes forward to indicate that they would not have actualized the idea without the ClimBeR partnership. Conclusion: Having the idea in mind could have inspired FACS to look for a relevant partner. Yet there is no explicit confirmation of this hypothesis.				

The rival hypothesis is that FACS were mostly responsible for the development of the *digital climate-credit scoring tool* – with less influence from ClimBeR. In any case, they already had a product idea. This testimony the *hoop* test; the confession of a product idea could have inspired interest in the innovation – a **necessary criterion**. However, there is **insufficient evidence** attributing the *digital climate-credit scoring tool* exclusively to the preconceived product idea. We therefore conclude that ClimBeR was mostly responsible for the development of the *digital climate-credit scoring tool*.

With respect to CSPDxCF, existing evidence shows that ClimBeR worked in partnership with **Interpeace** to develop the all-in-one solution for assessing conflict- sensitivity of climate finance. Our hypothesis is that ClimBeR played an influential role in development of the platform. While evidence shows that ClimBeR developed the innovation in partnership with **Interpeace**, the documented leadership role played by CGIAR programs with exclusive focus on climate science is evidence that passes the *smoking-gun* test. They provide **strong and sufficient** evidence in favour of the hypothesis that ClimBeR had great influence in the development of the platform. The evidence is, however, **not a necessary** criterion for confirming the hypothesis.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw-in-the-wind	Hoop	Smoking-gun	Double decisive
Digital climate-credit scoring tool	Main hypothesis: ClimBeR played an influential role in development of <i>CSPDxCF platform</i> .				
	Clue: CGIAR is confirmed as leading the process and having a climate science focus, thus giving credence to their influence. Conclusion: Documentation passes the <i>smoking-gun</i> test and therefore confirms the hypothesis.				
	Rival hypothesis: Interpeace was mostly responsible for development of <i>CSPDxCF platform</i> .				

Clue: Interpeace has a mandate for strengthening capacity for peacebuilding but has no express focus on climate risks as a threat multiplier. Conclusion: Straw in the wind evidence. Having a focus on peacebuilding is relevant, but neither necessary nor sufficient.				
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The rival hypothesis is that **Interpeace** played an influential role in the development of the innovation. While **Interpeace** works to “strengthen the capacities of societies to manage conflict in non-violent, non-coercive ways”, their efforts are geared towards “developing social and political cohesion”, without express focus on climate risk motivated conflicts. This evidence is therefore neither necessary nor sufficient to support the hypothesis. We therefore conclude that ClimBeR had greater influence on the development of the CSPDxCF.

We also assess the contribution of ClimBeR on the development of the *Climate Security Investment Plan* (CSIP) methodology. The document detailing the methodology accredits the development to CGIAR researchers. There is no indication of partnership in the development of the methodology. This evidence is therefore double decisive and confirms that ClimBeR was exclusively responsible for development of the CSIP methodology.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw-in-the-wind	Hoop	Smoking-gun	Double decisive
Digital climate-credit scoring tool	Main hypothesis: ClimBeR played an influential role in development of <i>CSIP</i> methodology.				
	Clue: Document describing the methodology was exclusively authored by CGAIR researchers – no partners mentioned. Conclusion: Evidence is <i>double decisive</i> and therefore confirm the hypothesis.				
	Rival hypothesis: Other partners were mostly responsible for development of <i>CSIP</i> methodology.				
	Clue: No one is mentioned as a partner in the existing documents detailing the methodology. Conclusion: Not applicable.				

In the absence of any evidence that could support any rival evidence, we conclude that ClimBeR was exclusively responsible for development of the CSIP methodology.

Gender Equity and Social Inclusion: The mainstreaming of gender considerations into ClimBeR workstreams has been achieved largely through the effort of the GESI team in ClimBeR. This has mainly been through the “voices of change”, mainstreaming of gender and social equity, and development of

indicators and data collection tools. The documentations and testimonials below are a proof of this contribution.

1. *Voices of change*. ClimBeR worked with partners to facilitate amplification local voices in climate adaptation policies and programming. According to CEMERIDE, one of the partners, the partnership “enhanced content and quality of **“voice of change”**, thus enhancing quality of advocacy work”.
2. *Social equity in building resilience to climate variability and extremes*. ClimBeR has developed a framework for mainstreaming gender and social equity considerations in climate adaptation and resilience programming. The document <https://hdl.handle.net/10568/125398>, was developed by ClimBeR in collaboration with the Nordic Africa Institute
3. *Gender and social equity indicators*. ClimBeR has also developed GSE indicators to guide the development of data collection tools and thus ensure that gender and social inclusion considerations are mainstreamed in climate adaptation and resilience studies. See <https://hdl.handle.net/10568/128662>.

The **significance of these contributions by ClimBeR** is assessed below.

For voices of change, we hypothesize that ClimBeR played an influential role in development and delivery of “voices of change”. According to CEMIRIDE, “Partnership with research enhanced content and quality of **“voice of change”**, thus enhancing quality of advocacy work”. These testimonies pass the *smoking-gun* test; they are **strong and sufficient** evidence in favour of the hypothesis that ClimBeR had great influence on the “voices of change” innovation. Nevertheless, they are not a necessary criterion for confirming the hypothesis.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw- in-the- wind	Hoop	Smoking -gun	Double decisive
Voices of change: increased participation by vulnerable groups in design of climate change adaptation	Main hypothesis: ClimBeR played an influential role in development and delivery of “voices of change”.				
	Clue: Partners indicate that Climber enhanced quality of “voices of change” through better story telling skills and research support.				
	Conclusion: Testimonies such as these are a confirmation that pass the <i>smoking-gun</i> test and therefore confirm the hypothesis.				
	Rival hypothesis: RECONCILE and CEMIRIDE were mostly responsible for development and delivery of “voices of change”.				
	Clue: The organization has local -presence and has climate change- and vulnerability- focused programs.				
	Conclusion: The focus on enhancing the voice of marginalized in climate change debate could have inspired them to work on “voices of change”, thus passing the hoop test. Yet there is no explicit confirmation of this hypothesis.				

The rival hypothesis is that partners – CEMIRIDE and RECONCILE were mostly responsible for the production and dissemination of the “voices of change” – with less influence from ClimBeR. The partners have climate change programs. CEMIRIDE for instance has a vision of “*making sure there is substantive involvement of marginalised indigenous communities in climate governance in Kenya*” <https://www.cemiride.org/climate-change/>. Accordingly, CEMIRIDE has a project entitled: “*Strengthening the Voices of Women with Disability to Actively Participate in Climate Change Policy and Negotiations (STREVOW)*”. This evidence passes the *hoop* test; the climate change focus of CEMIRIDE and RECONCILE and their focus on elevating voices of marginalized could have inspired interest in “voices of change” – a **necessary criterion**. However, there is **insufficient evidence** attributing “voices of change” exclusively to the focus of the two partners on climate change. While the hypotheses test does not exclusively attribute success of “voices of change” to ClimBeR, there is stronger evidence that ClimBeR had greater influence.

For gender mainstreaming, we hypothesize that ClimBeR had an overriding influence in the framing of social equity issues that are of concern to climate resilience building and development of indicators and data collection tools. The need for mainstreaming and integration of gender and social equity consideration in climate adaptation strategies is underscored was demonstrated at the inception of ClimBeR, as demonstrated in the proposal. Moreover, the 2 publications (<https://hdl.handle.net/10568/125398> and <https://hdl.handle.net/10568/128662>) that document the framing of social equity considerations and development of indicators, respectively, were produced with substantial leadership from ClimBeR. These pieces of evidence are *double decisive* - **necessary and sufficient evidence**, which therefore confirms that ClimBeR had an overriding influence in the two processes.

Significant change (output/outcome)	Hypotheses	Hypotheses tests			
		Straw-in-the-wind	Hoop	Smoking-gun	Double decisive
GSE framing, indicators, and data collection tools	Main hypothesis: ClimBeR has an overriding role in framing of GSE consideration and development of indicators and data collection tools.				
	Clue: Existing documents – proposal and technical briefs show that ClimBeR played a leading role in the work.				
	Conclusion: Documentation pass the <i>double decisive</i> test and therefore confirm the hypothesis.				
	Rival hypothesis: Other partners played a greater role in the framing of gender consideration and development of indicators and tools for data collection.				
	Clue: The Nordic AFRICA INSTITUTE contributed to the “GSE issues framing” document.				
	Conclusion: NORDIC AFRICA INSTITUTE contributed only for the GSE issue framing, even then leadership was still with ClimBeR still led the process.				

The rival hypothesis is that partners and other stakeholders were largely responsible for gender and social equity mainstreaming and development of indicators and tools for data collection. We, however,

find no evidence of the role of other partners other than the Nordic Africa Institute. Moreover, their contribution was limited to the briefing note on framing of GSE issues - <https://hdl.handle.net/10568/125398>. This is therefore a *straw in the wind* evidence that only affirms the relevance of the partners in development of this documents but does not confirm the overriding role of these partners.

3.3.4 Early impact assessment (household survey)

To assess the early impacts of ClimBeR intervention on the households, we use RCC data collected from households in Embu, one of the intervention areas. Notice that access to agricultural credit is considered an important avenue through which smallholder farmers manage production and marketing risk, particularly those related to climate uncertainties. Credit access enables smallholder farmers to purchase improved agricultural inputs such as climate-smart varieties and fertilisers, and to adopt yield-enhancing and resilience-building technologies such as irrigation (Hailu et al., 2014; Simtowe et al., 2009). Further, credit access can enable farmers to take advantage of high value markets through contract farming (Meyer et al., 2017). In the developing countries, however, market imperfections constrain supply and farmer utilization of agricultural credit (Marjit & Mishra, 2023).

Credit access is constrained by both supply-side and demand-side factors. On the supply-side, information asymmetry is perhaps the most important factor as it makes it risky and more expensive to serve smallholder farmers, making lenders either ration the quantity of credit, charge higher interest rate or impose collateral requirements (Farrin & Miranda, 2015). From the demand-side, the smallholder themselves may opt out of the credit market, either because of lack of collateral or fear of losing it (Boucher et al., 2008; Guiringer & Boucher, 2008). Other factors may include the transaction costs associated with cumbersome loan application procedures, long travel distances to the lenders and low financial literacy (Balana et al., 2022). The alternative, which is the informal market, is not accessible to all farmers and even the farmers that are well known to the lenders may be discouraged by the prohibitively higher rates of interest charged (Demont, 2022).

Evidence shows that women face more credit constraints than men. This is because of the additional social and economic constraints women face that may substantially reduce their willingness to participate in the credit market and even the amount they can access (Elahi et al., 2018; Reboul et al., 2021; Zins & Weill, 2016; Timu et al. 2023). This limits them to less risky, low-return farm investments with the consequence of low production and productivity, food security and resilience to climate shocks.

Under ClimBeR, bundling agricultural credit with insurance, popularly called risk-contingent credit (RCC), is seen as a way of ameliorating both supply-side and demand-side constraints to credit access and utilization. RCC aims to improve credit uptake and farm investment by reducing the risk of borrowing, and collateral and/or guarantor requirements (Mishra *et al.*, 2021; Ndegwa *et al.*, 2020; Timu et al. 2023). In this analysis, therefore, we explore whether or not this objective is being realized under the ClimBeR project.

Descriptive results show that there is a high correlation between crop insurance and agricultural credit, and that joint adoption of the two innovations is highly likely, and indication that insurance eases agricultural credit rationing (Table 6).

Table 6. Bivariate uptake of agricultural insurance and agricultural credit

Crop Insurance	Agricultural credit		Total
	Yes	No	
Yes	38%	0%	38%
No	6%	56%	62%
Total	44%	56%	100%

The rate of uptake of the two schemes and complementary services, and their correlations are summarised in Table 7.

Table 7. Rate of uptake of the schemes

Schemes and supportive services	
Agricultural insurance	39%
Agricultural credit	44%
Agro information/climate services	60%
Correlation between agricultural insurance and agricultural credit	0.87***
Correlation between agricultural credit and agro information/climate services	0.14***
Correlation between agricultural insurance and agro information/Climate services	0.15***

***, indicate that pairwise correlations are statistically significant at 1%

Of the smallholder farmers, 38% adopted both agricultural insurance and agricultural credit, 6% adopted only agricultural credit, 56% adopted neither insurance nor agricultural credit. It is also observable that no smallholder took up crop insurance without taking up agricultural credit. Of those who received credit, male farmers constituted 84% while the female farmers constituted only 16%. Notably, the smallholder farmers benefitting from the bundled insurance and credit received significantly higher amounts of credit relative to their counterparts receiving only credit (Table 8).

Table 8. Difference in credit amounts between beneficiaries and non-beneficiaries of RCC

	Holder of agricultural insurance		Difference
	Yes	No	
Average amount of credit received (KSh)	14637	3853	10784***

***, significant at 1%

Uptake of RCC is important for smallholder farmers for two main reasons:

1. It has led to uptake or enhanced use of improved, drought-tolerant crop varieties. This is either because the farmers are no longer worried of the risks associated with such varieties because of insurance cover or because the credit received is used to purchase the seeds which are usually too expensive for the households to purchase. For example, under the credit scheme, each smallholder farmer was entitled to up to 8 kgs of certified maize seed.
2. Uptake or increased use of fertilisers and pesticides, attributable to the borrowed funds. Each smallholder, under the credit scheme, was entitled to up to 100 kgs of planting fertiliser, 100 kgs of top-dressing fertiliser, 100 ml of foliar fertiliser and 100ml of pesticide, depending on individual farmer's own needs.

Notice also that the bundled insurance and agricultural credit comes with agronomic training of the farmers, exposing them to these improved production techniques. Consequently, under normal conditions, these farmers are more likely to realise increased production and productivity.

Previous studies have found that gender of the farmer, household size, land size, proximity to market and accessibility have been integral in influencing farm household production decisions (Maurice J Ogada *et al.*, 2014; Maurice Juma Ogada *et al.*, 2021). We, therefore, include these variables in the empirical model specification. Table 9 provides a summary of these variables and compares them across the households which received agricultural credit and those that did not in the reference period.

Table 9. Variables used in the bivariate analysis

Variable	Received Credit		Difference
	Yes	No	
Household size	5.2	4.6	0.6
Total land owned (acres)	3.6	3.9	-0.3
Distance to output market (walking minutes)	48.2	41.3	6.9
Distance to all weather road (walking minutes)	7.8	9.6	-1.8
Female farmers	16%	22%	
Male farmers	84%	78%	

Tests indicated that the variables were not statistically different between the credit recipients and the non-recipients. This shows that the sampling of the farm households was truly random, and results may be used to make generalisations for the study population. The bivariate probit regression results are shown in Table 10.

Table 10. Bivariate Probit Estimates of Uptake of agricultural insurance and agricultural credit

Variable	Insurance uptake		Credit uptake		Joint uptake
	Coefficient	Marginal effect	Coefficient	Marginal effect	Marginal effect
Gender of the farmer (0=female, 1=Male)	0.98*** (0.38)	0.30***	1.02*** (0.35)	0.33***	0.30***
Access to agro-information	0.40*** (0.141)	0.14***	0.37*** (0.139)	0.14***	0.14***
Household size	0.07* (0.04)	0.02*	0.05 (0.04)	0.020	0.02*
Total land owned (acres)	-0.01 (0.014)	-0.004	-0.02 (0.016)	-0.008	-0.005
Distance to output market	0.02*** (0.01)	0.01***	0.03*** (0.01)	0.01***	0.01***
Distance to mkt squared	-0.000003 (0.0000058)	-0.000001	-0.0001* (0.00006)	-0.000004*	-0.000002
Distance to all-weather road	-0.0005 (0.006)	-0.0002	0.003 (0.006)	0.001	0.0001
Gender*Distance to output market	-0.02*** (0.008)	-0.01***	-0.02** (0.008)	-0.01***	-0.01***
Intercept	-1.84*** (0.374)		-1.75*** (0.35)		
LR test (rho=0)	$\chi^2(1) = 95.7***$		$\chi^2(1) = 95.7***$		

Observation	354	354
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*, **, ***, indicate statistically significant at 10%, 5% and 1%, respectively.

Foremost, the significance of the rho justifies the use of bivariate probit. That is, the error terms of the insurance uptake and credit uptake equations are correlated and failure to take that into account in the analysis yields biased estimates. This is what bivariate probit, in our analysis, seeks to cure and, therefore, our results are consistent and reliable. Interpretation of these results, as usual, is based on the marginal effects.

The results show that the male farmers have 0.30 higher probability to take up agricultural insurance and/or both agricultural insurance and agricultural credit than their female counterparts. They also have 0.33 higher probability to take up agricultural credit than their female counterparts. Access to agricultural and climatic information services also increases the probability of uptake of the two schemes, either singly or jointly. Farm households with such access have 0.14 higher probability to take up agricultural insurance, agricultural credit or both agricultural credit and crop insurance simultaneously.

Households rely mainly on own labour for agricultural production. Therefore, it is not surprising that large household size is positively correlated with uptake of crop insurance scheme and joint uptake of crop insurance and agricultural credit. A rise of household size by one member increases the chance of uptake of crop insurance scheme or combined agricultural insurance and credit by 2%.

Distance from the farm to the output market had a non-linear effect on the uptake of the two schemes. Within some range uptake increases with the distance, whereby an increase of distance by one minute of walking time increases the probability of uptake of each of the schemes or their combination by 0.01. This is most probably because farm households very close to the market have alternative livelihood sources and may not invest much in agriculture while those far away from the market consider agriculture as the main source of livelihood. Beyond some range of distance, however, any further increase in distance to the output market only serves to diminish the likelihood of uptake of agricultural insurance and/or credit. This is understandable because if the market is too far, commercial production is discouraged and investment in agriculture is reduced.

Interaction of gender and distance to the output market shows that women farmers are more heavily affected by the distance than their male counterparts. The probability of male farmers getting rationed out of agricultural insurance and/or credit by incremental distance to the output market is 1% lower. Most probably this is because women have multiple tasks to perform at home that any activity that takes them too far from home would be less attractive. For example, if agricultural produce is for the market and the market happens to be too far, women farmers would be less attracted to farming, and this will reduce their access to agricultural insurance and/or credit.

Emerging issues in implementation of the bundled agricultural insurance and credit

The experimental scheme covers smallholder farmers for maize and beans on one acre. There is a deliberate effort to include women in the scheme. Training materials were gender-sensitive and during the training, women with young children were provided with childcare services. So far, 40% of the farmers covered are women.

Agricultural credit provided per acre is Ksh 22,000 in an e-voucher system, entitling a farmer to:

1. Eight kilogrammes of certified maize seed
2. One hundred kilogrammes of planting fertiliser
3. One hundred kilogrammes of top-dressing fertiliser
4. One hundred millilitres of foliar fertiliser
5. One hundred millilitres of pesticides

Each farmer, however, has the freedom to tailor the credit to his/her needs by selecting what he/she requires for his farm. The amount of insurance premium is Ksh 2000 per acre of maize-beans mixed system. Loan repayment is made after the crop harvest. Overall, the scheme has led to increased use of certified seeds, fertilisers, and pesticides. Some farmers have also increased land under cultivation.

The following issues were raised by the farmers:

1. The scheme was too restrictive, both in terms of acreage and the crops. They would have loved a scheme which gives them freedom on the size of land to insure and the type of crops to grow. Maize was, particularly noted not to do very well in some parts of the County. For example, farmers in Kiambere would have preferred green grams, cowpeas, and sorghum instead of maize and beans.
2. The credit given is in the form of e-voucher for specified inputs only. The farmer is not able to redeem the vouchers for labour, transportation, farm structures, storage, etc. There were also cases where farmers got notifications that they had redeemed certain vouchers when they had actually not done so.
3. There were instances where inputs were delayed and/or were inadequate (e.g. pesticides).

3.4 Reflection on ClimBeR products

There are many products/innovations associated with ClimBeR, most of which are public goods. There are, however, some that are attractive to private sector investments. These include:

1. RCC – is attractive to banks and insurance companies. In Embu for instance, Family Bank has provided bundled credit to 260 farmers, who have also been insured by APA. Equity Bank is also providing bundled credit to farmers in Machakos. Activities around RCC were well implemented but there are some challenges that could undermine interest, wider uptake, and further private sector investment. These include:
 - i. Delayed supply of relevant inputs and inappropriate inputs
 - ii. High interest rates and insurance premium
 - iii. Restriction on crop types and land area coverage
2. Climate information services – Climate information services have also attracted the interest of MEDIAE, who are serializing climate-related “Shamba Shape-Up” episodes. Collaboration with ClimBeR has also enhanced the quality and visibility of iShamba, thus making it more marketable to potential partners of MEDIAE.
3. Digital finance scoring tool – has the potential to be attractive to banks but interviews with LANDEX and ECLOF that would uncover this potential/achievement have not taken place.
4. Climate Security training module, although prepared for implementation by Jomo Kenyatta University, can be taken up by the private capacity developers and institutions of higher learning.

5. Other products such as iFEED, AWARE, LNOB indicators, CSO, climate security training modules and Climate-Smart Governance dashboard are more inclined towards public goods and would rather be taken up and funded by the government post the project; otherwise, they may meet challenges in terms of maintenance and updating.

We subjected these products to such evaluation criteria as sustainability, additionality, efficiency, effectiveness, coherence, relevance, impact and gender and social equity.

Sustainability

Innovations such as RCC provide significant **value proposition** to banks that benefit from increased clientele base while MEDIAE benefit from enhanced quality of education services (agroclimatic services) that is likely to attract for them more business partners. Innovations such as the training course on *CSA-peacebuilding nexus* have **strong alignment to the mandate** of JKUAT (partner organizations), thus providing opportunity for continuity. So is agroclimatic information services of ClimBeR to the education mandate of MEDIAE. Innovations such as iFEED **embed structures within existing institutions**, e.g., the TWG approach with some members who will double up as “champions of change”. Through their influencing role within respective agencies/organizations, they will continue to champion the outcomes of the iFEED process.

	Value proposition to partner organization	Alignment to mandate of partner organization	Institutional embeddedness
RCC			
CSA-CS training modules			
iShamba			
Shamba Shape Up (SSU)			
iFEED			

Note: Green colors indicate that outputs fulfil the evaluation criteria because of the reasons outlined in the columns

Additionality

Collaboration with ClimBeR brought onboard **research expertise** that the banks leveraged to develop the RCC product. The rigorous research and the associated flexibility to test innovations is a “luxury” that businesses usually do not have. MEDIAE also leveraged this expertise to unpack complex weather information from meteorology department into simpler agroclimatic information. They also leveraged research expertise to develop extension advisory that accompany agroclimatic information channeled through Shamba Shape-Up and iShamba. The CSA training at JKUAT benefits from the **new perspective** of climate security that exploits CSA as a tool to minimize the risk of climate change as a threat multiplier, thus adding value to their training program.

	Research capabilities	New/complementary perspective
RCC		
CSA-CS training modules		
iShamba		
SSU		

Note: Green colors indicate that outputs fulfil the evaluation criteria because of the reasons outlined in the columns

Efficiency

While RCC is costly to set up and leads to higher premiums, it provides an efficient way to mobilize smallholder farmers and enlist them into a commercial risk management program. The process of developing the training modules for the CSA-CS nexus is, however, cheap compared to standard curriculum development that could be costly in terms of the cost of subject matter specialists. Moreover, the process is effective as it enables achievement of two goals: adaptation and climate security through a single course. The collaboration enabled MEDIAE to access research expertise that they would otherwise have had to pay for. Being a mass media outlet, SSU and iShamba enable wider outreach at relatively low cost.

	Cost	Process
RCC		
CSA-CS training modules		
iShamba		
SSU		

Note: Green colors indicate that outputs fulfil the evaluation criteria because of the reasons outlined in the columns while red indicates otherwise

Effectiveness

RCC has given target farmers the confidence to invest in climate adaptation technologies, thus has been more effective in enabling uptake of appropriate practices. On the other hand, the *CSA-CS nexus* course combines CSA and climate security thus leveraging one course to achieve adaptation and climate security. Mainstreaming CS into standard curriculum at JKUAT has also improved effectiveness of the university training, making the CSA course more robust. Enhanced **agroclimatic information services have also been mainstreamed into iShamba and SSU platforms** thus making the respective platforms more effective in facilitating behavior change. The platforms **leverage research expertise** to synthesize complex weather data into simpler agroclimatic information, whose effectiveness in catalyzing change in practices is further enhanced through the accompanying agro-advisory.

	Influencing behavior	Process (mainstreaming)
RCC		
CSA-CS training modules		
iShamba		
SSU		

Note: Green colors indicate that outputs fulfil the evaluation criteria because of the reasons outlined in the columns

Coherence

Provision of bundled financial products through RCC is consistent with government policy of agricultural risk management and access to finance by farmers. The training course on the nexus of CSA and climate security is also aligned with the government's intention to minimize insecurity in climate hotspots. Finally, provision of agroclimatic information through SSU and iShamba platforms contributes to the government agenda of informing the population on the risks of climate change and appropriate mitigation and adaptation measures. Agro-advisory that accompany weather information also contributed to the national government's agenda of agricultural risk management.

	Risk management & access to finance	Insecurity	Information provision
RCC			
CSA-CS training modules			

iShamba			
SSU			

Note: Green colors indicate that outputs fulfil the evaluation criteria because of the reasons outlined in the columns

Relevance

RCC addresses the needs and limitations of smallholder farmers – agricultural risks and limited access to finance. The training course, on the other hand, aligns with the training mandate of JKUAT and specifically with their curriculum on climate smart agriculture. Similarly, agroclimatic information services are aligned with MEDIAE's focus on climate change and adaptation as well as the educational role of the institution. These services are also relevant to the information asymmetry that characterizes smallholder farming.

	Smallholder challenges	Partner mandate
RCC		
CSA-CS training modules		
iShamba		
SSU		

Note: Green colors indicate that outputs fulfil the evaluation criteria because of the reasons outlined in the columns

Impact

Adoption of RCC is expected to enhance uptake of climate smart agricultural technologies and practices, leading to increased productivity. The training course is expected to deliver more informed graduates with appropriate skills to promote use of CSA as a tool for peacebuilding. Similarly, farmers who receive agroclimatic information services are expected to make better decisions and realize improvement in yields.

	Knowledge	Practices	Yields
RCC			
CSA-CS training modules			
iShamba			
SSU			

Note: Green colors indicate that outputs fulfil the evaluation criteria because of the reasons outlined in the columns

Gender and social equity

Women were deliberately targeted in the RCC product while the training course factors in gender considerations. The SSU and iShamba services target all farmers irrespective of gender but the call center has female staff that target women farmers.

	Targeting	Gender responsiveness
RCC		
CSA-CS training modules		
iShamba		
SSU		

Note: Green colors indicate that outputs fulfil the evaluation criteria because of the reasons outlined in the columns

4. Conclusion

Tremendous progress has been made across Work Packages, with some activities showing indications of early outcomes. Some farmers are already adopting bundled financial products, while a sizeable number are also accessing climate information services accompanied by financial literacy and agronomic training. ClimBeR partners such as IGAD and the national government in Kenya are also integrating climate security in resilience programming – IGAD, through their strategy and national government, through NCCAP III. Moreover, ClimBeR has developed the Climate Security Index to enhance measurement and monitoring of climate security vulnerability, which has already been applied to measure climate security vulnerability in Kenya.

ClimBeR has also achieved a milestone in amplifying the voices of marginalized and “most at risk” groups in the climate change and adaptation debate, ensuring that the interests of these groups are at the core of designing adaptation solutions. A notable achievement in this respect has been the participation of these “Voices of Change” at COP 27 and 28. Through the CSO, ClimBeR has also identified climate security hotspots in the country and the vulnerable groups in respective hotspots and outlined solutions for reducing the “threat multiplier” effect of climate change. Towards enhancing policy efficacy on climate security, CimBeR undertook an assessment of climate-, agriculture-, and peace-related policies and strategy documents to ensure coherence in climate policies, thus setting the stage for possible review of these policies to enhance coherence of climate security policies.

At the output level, ClimBeR has built consensus on the concept of climate security, including defining pathways through which climate change exacerbates conflict. ClimBeR has also worked to embed climate security at the core of climate change adaptation measures, by developing a training course on *“Climate, Peace, and Security in the Agriculture Sector”*. This course will ensure that practitioners promote CSA as a tool for minimizing climate security risks. The course is intended to enhance understanding of climate security, unpack the role of climate action and peace in promoting inclusivity, outline the contribution of CSA to peacebuilding, and identify tools to support climate security interventions.

Towards a climate resilient, socially equitable, and nutrition-secure future, ClimBeR has promoted iFEED – an integrated assessment framework for identifying equity-sensitive, nutrition secure pathways to resilience against climate change. So far, nutrition indicators for iFEED modelling have been identified. A technical working group has also been formed to steer the process of unpacking the outcome of iFEED modelling and to champion the implementation of policy actions that will emerge. Activities under this Work Package appear to be behind schedule and urgent action is needed to ensure timely delivery of obligations.

These achievements have been realized courtesy of diverse partnerships across the Work Packages. Most of these partners had mandates that aligned well with the goal of ClimBeR and therefore made substantial contributions. We, however, note that across board, ClimBeR had greater influence on the recorded achievements and was overwhelmingly responsible for the progress made.

Rigorous assessment of existing evidence shows that while risk management of clients remained on the radar of Equity and Family banks, collaboration with ClimBeR was largely responsible for the rollout of the RCC. On the other hand, our analysis shows that ClimBeR and MEDIAE leveraged each other’s

competencies and expertise to jointly deliver the agroclimatic information services. The two were therefore equally responsible for the realized achievement. This is a clear demonstration of a perfect match between ClimBeR and MEDIAE.

On gender and socially inclusive pathways, we find that the success of “voices of change” cannot **exclusively** be attributed to ClimBeR. However, there is strong evidence that ClimBeR played a great role in amplifying the voices of vulnerable communities at the table of climate change policy discussions. Moreover, ClimBeR was solely responsible for the development of the climate security policy coherence toolkit. Moreover, the assessment of diverse policy and strategy documents was conducted exclusively by ClimBeR.

ClimBeR also made substantial contribution to integration of climate security into resilience programming. Notable contributions include integration of climate security into IGAD strategy and NCCAP III in which ClimBeR is acknowledged as having been responsible for inclusion of climate security into respective resilience programming documents. Indeed, previous versions of respective documents did not include climate security. This gives credence to the assertion that participation of ClimBeR in the development of these documents was central to the consequent integration of climate security. The development of the training course on “*integrating climate change adaptation with peacebuilding*” is also largely attributed to ClimBeR.

Towards improved governance for adaptive transformation, we find that the development of LNOB indicators for ClimBeR initiative was largely achieved by the core ClimBeR team with limited if any contribution from other organizations. We also find strong evidence that ClimBeR had greater influence on unpacking the concept of MPG, including demonstrating through the study, the importance of local leadership in execution of community adaptation interventions. The evidence, however, does not invalidate the potential contribution of partner organizations such as Water Integrity Network (WIN).

Overall, we find substantial progress in the implementation of activities that were planned by the ClimBeR initiative. There are strong indications that the realized outputs were well on course to delivering the intended outcomes, with early results indicating uptake of various innovations developed by ClimBeR. We also find strong evidence of greater influence of ClimBeR in realization of the significant milestones. Much of this progress has been realized under Work Package 1 and 2. Despite lagging slightly behind, activities planned under Work Package 3 and 4 also need to be expedited to enhance the development of planned outputs and realization of intended outcomes especially with respect to *Early Warning, Early Action, Early Finance*” (AWARE) platform and *Climate Smart Governance dashboard*.

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